



ANTEO+: A subchannel code for thermal-hydraulic analysis of liquid metal cooled systems



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HIGHLIGHTS

- The code structure is presented in detail.
- The performed validation is outlined.
- Results are critically discussed assessing code accuracy.
- Conclusions are drawn and ground for future work identified.

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ABSTRACT

Liquid metal cooled fast reactors are promising options for achieving the high degrees of safety and sustainability demanded by the Generation IV paradigm. Among the critical aspects to be addressed in the design process, thermal-hydraulics is one of the most challenging; in order to embed safety in the core conceptualization, these aspects are to be considered at the very beginning of the design process, and translated in a design perspective. For achieving these objectives the subchannel code ANTEO+ has been conceived, able to simulate pin bundle arrangements cooled by liquid metals. The main purposes of ANTEO+ are simplifying the problem description maintaining the required accuracy, enabling a more transparent interface with the user, and having a clear and identifiable application domain, in order to help the user interpreting the results and, mostly, defining their confidence. Since ANTEO+ relies on empirical correlations, the validation phase is of paramount importance along with a clear discussion on the simplifications adopted in modeling the conservation equations. In the present work a detailed description of ANTEO+ structure is given along with a thorough validation of the main models implemented for flow split, pressure drops and subchannel temperatures. The analysis confirmed the ability of ANTEO+ in reproducing experimental data in its anticipated validity domain, with a relatively high degree of accuracy when compared to other classical subchannel tools like ENERGY-II, COBRA-IV-I-MIT and BRS-TVS.

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1. Introduction

Generation IV concepts, envisaging innovative nuclear reactor's design features, are on the research spotlights because of their ambitious vision, which has safety by design and sustainability as key aspects. The former pushes safety related considerations deeper in the design process including them since the conceptualization stage. Sustainability, which can be translated in an efficient

exploitation of uranium resources and a drastic simplification in waste management, could be attained with the closure of the fuel cycle for which fast reactors (FR) are the most credible option. Among the FR community the most studied concepts involve liquid metal coolants (LMFR), like sodium, lead–bismuth eutectic and lead, thanks to their favorable neutronic and thermal-hydraulic characteristics (Lazaro et al., 2014; Sarotto et al., 2013; Grasso et al., 2014).

The high flux and high temperature combination typical of a LMFR requires, for safety reasons, an accurate knowledge of the thermal field inside the sub-assembly (SA). In line with the Generation IV philosophy, safety is deeply embedded in the design phase, requiring that thermal-hydraulic conditions must be quantitatively assessed at the beginning of the aforementioned phase.

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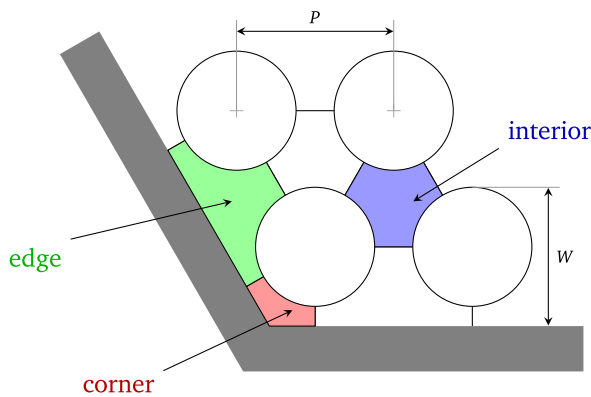


Fig. 1. Subchannel definition for an hexagonal assembly (rods in a triangular lattice).

For these reasons and others (e.g. optimization) various numerical techniques had been conceived over the years starting with 1D approaches down to detailed CFD simulations. For preliminary design verification purposes however the dominant approach has been the so called subchannel (SC) method (Sorokin et al., 2000), where conservation equations are integrated over a channel, representing the flow area between fuel pins or between pins and duct, if present (Fig. 1). This integration or “lumping” procedure assumes constant physical variables (i.e. temperature, pressure and velocity) across the control volume while energy and momentum exchange phenomena between SCs are modeled with empirical correlations.

The SC method was selected because its overwhelming history of application to nuclear reactors which, following Sorokin et al. (2000), started in the early 1960s with codes like THINC-1 (Zernick et al., 1962) in the USA, JOTO (Okamoto et al., 1970) in Japan and TEMP (Zhukov et al., 1973) in USSR. They enabled heterogeneous power production and geometry to be considered, along with inter-channel mixing, during the evaluation of the hot spot location. Subsequently, in the 1970s, codes simulating the presence of a wire spacer like COTEC (Novendstern, 1972) SWEEP (Ginsberg, 1972) and TEMP-M (Bogolovskaya et al., 1983) were developed, mainly relying on “force functions” models based on a continuity approach and so not extendable to mixed or natural convection conditions (Ninokata et al., 1987). All these tools assumed a constant pressure distribution over the SA cross section and so they were not suited for simulating large density excursions situations like bulk coolant boiling or flow blockage. In order to tackle transient conditions and transverse pressure imbalances, codes like COBRA II (Rowe, 1970), FLICA III (Plas, 1974), CORTRAN (Khan et al., 1979) and many others were developed. In the same period another family of codes like SUPERENERGY (Wolf et al., 1987) and MIF-C (Martchuk, 1973) were emerging accounting for inter-assembly thermal effects which were demonstrated to perturb several outer rows of pins. Finally in the 1980s the SC method was extended to flow blockage situations and coolant boiling with codes like COBRA-IV-I (Wheeler, 1976), TOPFRES (Arai and Hirata, 1984), TEMP-MIF (Kazachkovsi et al., 1985), SABRE-3B (Macdougall and Lillington, 1984) and others, entering the field of safety analysis. Some of these codes families were originally intended for water applications and then branched to liquid metal coolants (LMCs), mostly sodium (e.g. COBRA), while others were tailored, since the beginning, for FRs (e.g. SUPERENERGY). Recently, a number of codes has been converted for LMFRs simulation like MATRA-LMR (Kim et al., 2002), MATRA-LMR-FB (Ha et al., 2009), SACOS-PB (Wang et al., 2013), COBRA-IV-I-MIT (Fricano, 2012) and COBRA-LM (Liu and Scarpelli, 2015) or were specifically envisaged for heavy metals like BRS-TVS (Afremov et al., 2004), but none of them has been thoroughly validated with experimental data and their accuracy and validity domain remains somewhat unclear.

An alternative path has been taken with the steady state SC tool called ANTEO (optimized thermal-hydraulic analysis from the Italian **AN**alisi **TER**moidraulica **OTT**imizzata) (Cevolani, 1996) with the main objective of having a clear and identifiable application domain and thus helping the user in the results interpretation phase necessary in a rational design process. To fully exploit this idea, the validity domain has been decided beforehand and then a coherent set of equations has been derived. This is the reverse of the usual approach historically followed by the SC codes just reviewed (COBRA-like codes being a significant example) that starts from the definition of a set of equations, aiming to model a quite general set of physical phenomena with the highest degree of completeness and let their specific validity and effective application domain be defined only after, in real world tests. The path chosen in ANTEO allows simplifications to be performed where possible, compatibly with the intended application domain, without significantly penalizing accuracy, thereby obtaining a double advantage: firstly it is easier for the user to understand the calculation flow and secondly there is a considerable speed up in the simulation avoiding any convergence problem.

ANTEO was initially conceived for light water simulations and has been recently converted to lead in a branch version called ANTEO-LFR (Lead Fast Rreactors) (Mattioli and Cevolani, 2012), which has been used for the SA thermal-hydraulic characterization of the ALFRED (Advanced LFR European Demonstrator) core design (Grasso et al., 2014). A generalized version of ANTEO-LFR, applicable to other liquid metals like sodium and lead–bismuth, called ANTEO+ has been now developed, with improved modeling and geometrical capabilities encompassing bare, gridded and wire spaced bundles.

Because SC codes rely mainly on empirical correlations, and because one of ANTEO+ main objectives is to have a well defined application domain, the validation phase is of paramount importance for establishing the degree of confidence in the results along with a clear statement of the simplifications adopted in modeling the conservation equations. For these reasons, the founding assumptions behind ANTEO+ are analyzed and a thorough validation is presented encompassing all the main models, cross checking also that the anticipated and actual validity domains coincide, thus confirming the correctness of the derived set of equations and approximations.

A comprehensive description of ANTEO+ structure is presented in Section 2, stressing the expected range of application. Next the code validation for both pressure and temperature predictions is discussed comparing ANTEO+ also with other codes in order to better contextualize the targeted accuracy, meaning, in the present work, the ability to reproduce experimental data. In Section 4 an analysis of the results follows and finally in Section 5 conclusions are drawn and further desired improvements are suggested.

2. ANTEO+

Inheriting the same approach of its preceding versions, ANTEO+ main objective is to simplify the problem description without penalizing accuracy thus enabling a more transparent interface with the user and, at the same time, considerably speeding up the calculation. For achieving this double task, at first the various terms of the governing equations must be analyzed, identifying those that do not contribute significantly to the particular problem: obviously, deciding to discard a particular phenomenon will restrict the validity range of the code, so the field of application must be decided beforehand and, then, a set of equations and approximations coherently set up.

For ANTEO and ANTEO+ (from here on only ANTEO+ will be explicitly nominated, although ANTEO solves the same set of

equations) the steady state, single phase, forced convection regime was selected because is the one usually of interest in the preliminary design phase of a power reactor.

In the following each steady state constitutive equation is analyzed and the adopted simplifications highlighted and discussed.

2.1. Mass conservation

The mass conservation equation for a SC reads

$$\frac{d\dot{m}_i}{dz} = - \sum_{j=1}^J W_{ij}, \quad (1)$$

where $\dot{m}_i$ is the mass flow rate of the i th SC, z is the axial coordinate, J is number of SCs adjacent to i and W_{ij} is the transverse mass flow rate per unit length between SCs i and j , which accounts for diversion cross flow due to radial pressure differences, usually induced by boiling, density gradients or geometric variations. In single phase forced flow, radial pressure and density gradients are negligible because the flow has a strong axial character, and so the diversion term can be eliminated from Eq. (1). The resulting mass conservation equation in ANTEO+ is then

$$\dot{m}_i = \text{const}, \quad (2)$$

where SCs do not exchange mass with their neighboring ones.

2.2. Axial momentum conservation

The axial momentum conservation equation can be expressed as

$$\begin{aligned} \frac{d(P_i A_i)}{dz} = & - \frac{d(\dot{m}_i v_i)}{dz} - A_i \rho_i g - \frac{1}{2} \rho_i v_i^2 A_i \frac{f_i}{D_{Hi}} - A_i \Delta P_{\text{form},i} \\ & - \sum_{j=1}^J W_{ij} v^* - \sum_{j=1}^J W_{ij}^M (v_i - v_j), \end{aligned} \quad (3)$$

where P_i is the pressure of the i th SC, A_i is the flow area, v_i is the coolant axial speed in the SC, ρ_i is the SC density, g is the gravitational acceleration, f_i is the Darcy friction factor with the walls surrounding the i th SC, D_{Hi} is the hydraulic diameter defined as $D_{Hi} = 4A_i/p_{wi}$ with p_{wi} indicating the wetted perimeter, $\Delta P_{\text{form},i}$ is the form pressure loss per unit length due to changes in flow direction or geometry like the ones in the presence of spacers or at the inlet and outlet of the fuel bundle, v^* is the effective speed transported by cross flow and W_{ij}^M is the effective mass exchange rate between SCs i and j for momentum transfer per unit length. The last term is labeled as *effective* because it is a fictitious mass flow defined in order to preserve the turbulent momentum exchange between SCs; in single phase, indeed, turbulent mixing does not result in a net exchange of mass, even if, there is a net exchange of momentum and energy.

The diversion term is neglected as in Eq. (1); W_{ij}^M can be eliminated from Eq. (3) because in forced convection the SCs are weakly hydro-dynamically connected and, if necessary, this effect can be partially accounted for in the flow split model. Finally the SC index on the pressure can be dropped because it is uniform on the SA cross section as we shall see (Eq. (6)). Eq. (3) can be then simplified as

$$\frac{d(PA_i)}{dz} = - \dot{m}_i^2 \frac{d}{dz} \left(\frac{1}{\rho_i A_i} \right) - A_i \rho_i g - \frac{1}{2} \rho_i v_i^2 A_i \frac{f_i}{D_{Hi}} - A_i \Delta P_{\text{form},i}. \quad (4)$$

2.3. Transverse momentum conservation

The transverse momentum equation can be written as

$$\frac{\partial(v^* W_{ij})}{\partial z} + \frac{\partial(u W_{ij})}{\partial x} = (P_i - P_j) \frac{s}{\eta} - K_{ij} \frac{W_{ij}^2}{2\rho_i s \eta}, \quad (5)$$

where x is the transverse flow direction, u is the speed in the direction x , s is the clearance between fuel rods, η is a characteristic distance between SCs (i.e. centroid-to-centroid distance) and K_{ij} is a coefficient accounting for both frictional and form losses for the crossflow. Because W_{ij} is assumed zero the pressure is uniform on the SA cross section and the transverse momentum equation simply states that

$$P_i = P_j \quad \forall i, j. \quad (6)$$

2.4. Energy conservation

The energy equation has the following form

$$\begin{aligned} \frac{d(\dot{m}_i h_i)}{dz} = & \chi_i - \sum_{j=1}^J W_{ij} h^* - \sum_{j=1}^J W_{ij}^H (h_i - h_j) \\ & - \sum_{j=1}^J \rho_{ij} \alpha_{ij} \left(\frac{s}{\delta} \right) (h_i - h_j), \end{aligned} \quad (7)$$

where h_i is the enthalpy of the i th SC, χ_i is the linear power discharged to SC i from the surrounding fuel pins, h^* and W_{ij}^H are the energy counterpart of v^* and W_{ij}^M , ρ_{ij} is the average coolant density of SCs i and j , α_{ij} is the average thermal diffusivity and δ is an effective mixing length usually taken as the ratio of η and the conduction shape factor κ (Cheng, 1984). Contrarily to W_{ij}^M in the momentum equation, W_{ij}^H is important even in forced convection and must be accounted for. Eliminating the diversion term and considering Eq. (2) we arrive at

$$\dot{m}_i \frac{dh_i}{dz} = \chi_i - \sum_{j=1}^J W_{ij}^H (h_i - h_j) - \sum_{j=1}^J \rho_{ij} \alpha_{ij} \left(\frac{s}{\delta} \right) (h_i - h_j). \quad (8)$$

The term W_{ij}^H can also be properly enhanced to model flow sweeping due to a wire wrap spacer or flow scattering due to grid spacers.

The flow rate in a channel cannot be computed from the above system of equations, namely, (2), (4), (6), (8), but it must be calculated before starting the actual solution; a flow split model is therefore necessary in order to define the inlet conditions of every SC in the bundle.

As can be seen, the final set of equations solved by ANTEO+ is remarkably different from the one usually solved by COBRA-like codes, being more simple, avoiding to solve the transverse momentum equation and thus avoiding convergence problems with the possibility of maximizing modeling efficiency for the selected validity range.

Summarizing, the modeling adopted in ANTEO+ corresponds to a system of SCs dynamically connected only at the inlet and energetically connected throughout the whole simulated length.

2.5. Coolant temperature calculation scheme

The simplifications highlighted lead to a decoupling of the energy and momentum equations, now only related through the temperature dependency of the physical properties of the coolant. The decoupling considerably simplifies the numerical solution scheme because, once the inlet conditions are known from the

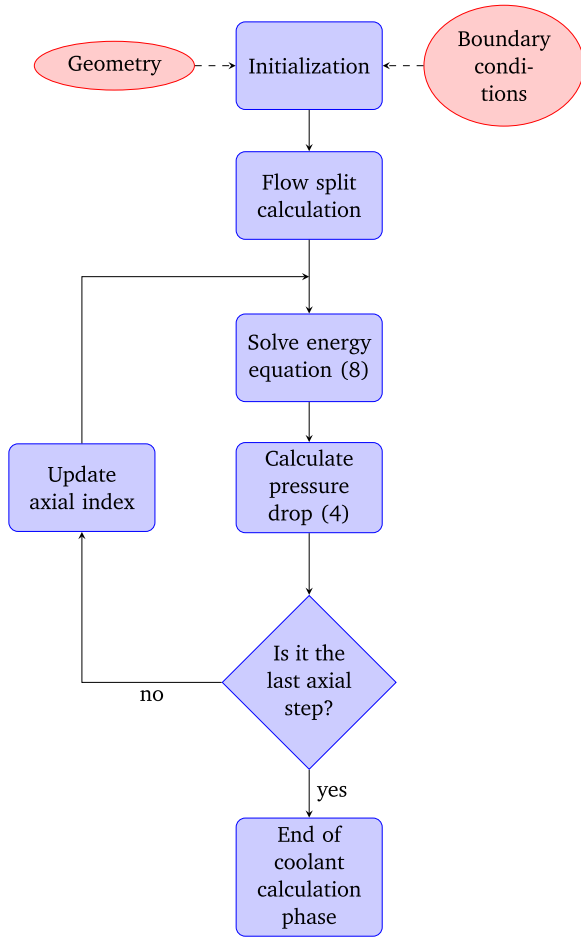


Fig. 2. Flow diagram of the coolant temperature calculation scheme in ANTEO+.

flow split model, the energy equation can be solved at a particular axial location with the Gauss elimination method, the pressure drop computed from (4) and then the sequence repeated for the next axial step. The above procedure is also depicted in Fig. 2.

The solution computed at an axial location is based only on the upstream nodes and does not depend on the subsequent ones, consequently the scheme is a forward one, coherently with the parabolic nature of the energy equation; this means that situations with a strong axial flow redistribution, like flow blockage, cannot be simulated. This is consistent with the forced convection assumption.

2.6. Fuel pin temperature distribution

Once the coolant temperature distribution is known, the internal temperature distribution of the fuel pin can be calculated (Fig. 3). ANTEO+ assumes unirradiated conditions, cold geometry and the gap filled with helium so the maximum calculated fuel temperature is likely to be conservative (Waltar et al., 2013).

2.6.1. Outer cladding temperature

The outer cladding temperature T_{co} is calculated at each axial step as

$$T_{co}(z) = T_b(z) + \frac{\chi_p(z)}{\pi D \alpha_p}, \quad (9)$$

where T_b is the maximum bulk coolant temperature surrounding the pin under analysis, χ_p is the linear power of the pin, D is the pin diameter and α_p is the convection coefficient.

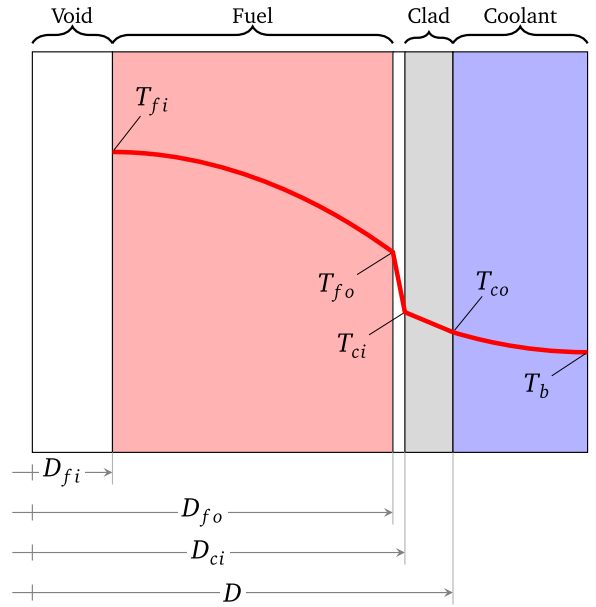


Fig. 3. Typical temperature distribution in a pin with hollowed fuel pellet and open gap.

2.6.2. Inner clad temperature

The inner clad temperature T_{ci} is computed as

$$T_{ci}(z) = T_{co}(z) + \frac{\chi_p(z)}{2\pi k_c} \ln \left(\frac{D}{D_{ci}} \right), \quad (10)$$

where k_c is the clad thermal conductivity and D_{ci} is the inner clad diameter. The thermal conductivity depends on temperature so (10) is effectively a non-linear equation which is solved with the method of successive approximations.

2.6.3. Outer fuel temperature

The calculation of the outer fuel temperature T_{fo} depends on the gap conductance, which in our case is proportional to the inverse of the helium thermal conductivity k_g . The equation is therefore

$$T_{fo}(z) = T_{ci}(z) + \frac{\chi_p(z)}{2\pi k_g} \ln \left(\frac{D_{ci}}{D_{fo}} \right), \quad (11)$$

where D_{fo} is the outer fuel diameter and is also solved with the method of successive approximations.

2.6.4. Inner fuel temperature

For computing the inner fuel temperature (or central temperature if there is no hole) T_{fi} the following equation is solved

$$\frac{1}{r} \frac{d}{dr} \left(k_f(T) r \frac{dT}{dr} \right) + Q_p = 0, \quad (12)$$

where k_f is the fuel thermal conductivity and Q_p is the power density. The boundary conditions are the known T_{fo} and zero heat flux at the center of the fuel or at the inner hole surface. Eq. (12) is discretized with finite difference and solved with the Gauss elimination method. Because k_f depends on T a Picard iteration is performed in order to converge on both variables.

2.7. State equations

For practically solving the momentum and energy equations on coolant side, and the heat equations for the fuel pin, a set of state equations is needed for the physical properties of the coolant, clad, gap and fuel. The selected values are reported in Tables 1–3, where

Table 1

Correlations for coolant physical properties implemented in ANTEO+; validity range is also included.

	Unit	Correlation	Validity range [K]
Sodium (Sobolev, 2011)			
ρ	kg/m ³	$1014.0 - 0.235 \cdot T$	371–1155
c_p	J/(kg K)	$1658.0 - 0.8479 \cdot T + 4.454 \times 10^{-4} \cdot T^2 - 3.001 \times 10^6/T^2$	371–1155
k_{cool}	W/(m K)	$104.0 + 4.7 \times 10^{-2} \cdot T$	371–1155
μ	kg/(m s)	$\exp(5.56835 \times 10^2/T - 6.4406)/(T^{0.3958})$	371–1155
Lead (OECD/NEA, 2015)			
ρ	kg/m ³	$11441.0 - 1.2795 \cdot T$	601–1903
c_p	J/(kg K)	$176.2 - 4.923 \times 10^{-2} \cdot T + 1.544 \times 10^{-5} \cdot T^2 - 1.524 \times 10^6/T^2$	600–1300
k_{cool}	W/(m K)	$9.2 + 1.1 \times 10^{-2} \cdot T$	600–1300
μ	kg/(m s)	$4.55 \times 10^{-4} \cdot \exp(1.069 \times 10^3/T)$	600–1473
Lead–Bismuth (OECD/NEA, 2015)			
ρ	kg/m ³	$11065.0 - 1.293 \cdot T$	403–1300
c_p	J/(kg K)	$164.8 - 3.94 \times 10^{-2} \cdot T + 1.25 \times 10^{-5} \cdot T^2 - 4.56 \times 10^5/T^2$	400–1100
k_{cool}	W/(m K)	$3.284 + 1.617 \times 10^{-2} \cdot T - 2.305 \times 10^{-6} \cdot T^2$	400–1100
μ	kg/(m s)	$4.94 \times 10^{-4} \cdot \exp(7.541 \times 10^2/T)$	400–1173

k_{cool} is the coolant thermal conductivity. All the properties are function of temperature only, so even if pressure losses are calculated by the code, the local pressure does not influence the physical parameters. For every region all the options currently available in ANTEO+ are listed. The material and state equations are identical for cladding and wrapper.

With respect to the fuel the only physical property is the thermal conductivity; three options are currently available.

2.7.1. UO₂

For uranium oxide the thermal conductivity is correlated as (Popov et al., 2000)

$$k_f = k_{f0} \cdot FD \cdot FP \cdot FM \cdot FRa,$$

$$k_{f0} = \frac{115.8}{7.5408 + 17.692 \cdot (T/1000) + 3.6142 \cdot (T/1000)^2} + 7.4105 \times 10^3 \cdot \left(\frac{T}{1000}\right)^{-2.5} \cdot \exp\left(-\frac{1.635 \times 10^4}{T}\right), \quad (13)$$

$$FD = 1,$$

$$FP = 1,$$

$$FM = \frac{1 - Po}{1 + 2 \cdot Po},$$

$$FRa = 1 - \frac{0.2}{1 + \exp((T - 900)/80)},$$

where k_{f0} is the thermal conductivity for fully dense unirradiated fuel, FD is the correction factor for dissolved fission products, FP is the correction factor for precipitated fission products, FM accounts for porosity, Po is the porosity fraction of the fuel and FRa accounts for radiation effects. FD and FP are taken as unity because for ANTEO+ the fuel is fresh.¹ As declared by the authors in Popov et al. (2000) the uncertainty on k_{f0} is 10% between 298 K and 2000 K and then raises to 20% between 2000 K and 3120 K due to higher scattering of experimental data in this range. The uncertainty on k_f is bound to be more.

The term FRa accounts for metallic fission products precipitates which behave differently in the cold and hot part of the fuel pellet. Specifically, in the cold part they are confined in the fuel matrix and contribute to phonon scattering while in the hot region they are mostly found outside the fluorite matrix (Lucuta et al., 1996).

¹ From input, the user can select a burnup and the code will calculate the appropriate value of FD and FP , the corresponding gap thickness must however be supplied by the user, knowing that for ANTEO+ the filling gas is always pure helium. ANTEO+ is not a fuel performance code, for accurate predictions of the fuel irradiation evolution the use of other codes is advised.

This term dictates the transition between the two regions, and is not burnup dependent, so it can eventually be omitted at zero burnup.

2.7.2. MOX

For mixed oxide fuels two correlations are available, the first taken from Popov et al. (2000) has the same structure as (13), namely

$$k_f = k_{f0} \cdot FD \cdot FP \cdot FM \cdot FRa,$$

$$k_{f0} = 2.3434 \times 10^{11} \cdot (T^{-2.5}) \cdot \exp\left(-\frac{1.635 \times 10^4}{T}\right) + \frac{1.1579}{A + C \cdot T},$$

$$A = 2.85 \cdot x + 3.5 \times 10^{-2},$$

$$C = (2.86 - 7.15 \cdot x) \times 10^4,$$

$$FD = 1,$$

$$FP = 1,$$

$$FM = \frac{1 - Po}{1 + 2 \cdot Po},$$

$$FRa = 1 - \frac{0.2}{1 + \exp((T - 900)/80)}, \quad (14)$$

where x is the deviation from stoichiometry. The validity range is 700–3100 K and a plutonium content in the range 3–15% with an uncertainty on k_{f0} estimated at 7% between 700 K and 1800 K raising to 20% between 1800 K and 3100 K. As before the uncertainty on k_f is bound to be higher. It is worth to notice that the typical plutonium content of a FR can exceed the validity range of the correlation; from Philipponneau (1996), however, seems that the effect of plutonium content is weak in the range 15% and 30% and so extrapolation should produce satisfactory results.

The second correlation is the one recommended by Philipponneau (1996) and has the following structure

$$k_f = \left(\frac{1}{A + B \cdot T} + C \cdot T^3\right) \cdot FC,$$

$$FC = \frac{1}{0.864} \cdot \frac{1 - Po}{1 + 2 \cdot Po},$$

$$A = 1.528 \cdot \sqrt{x + 9.31 \times 10^{-3}} - 0.1055,$$

$$B = 2.885 \times 10^{-4},$$

$$C = 7.638 \times 10^{-11}, \quad (15)$$

valid from 500 K up to the melting point. No uncertainty estimation was found in literature.

Table 2

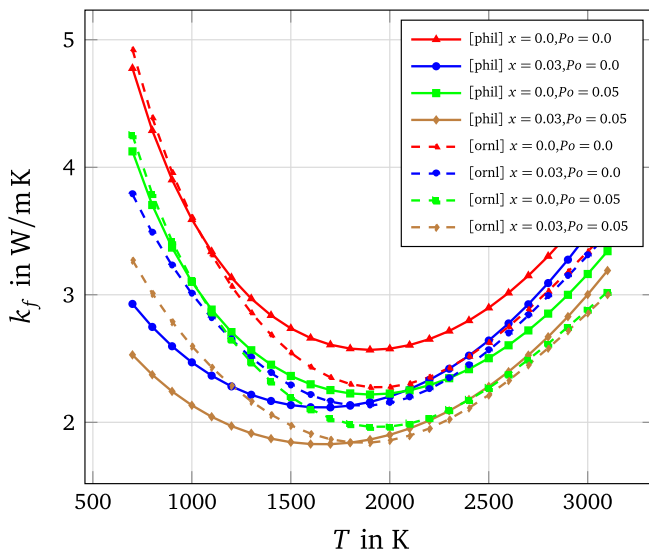
Correlations for cladding physical properties implemented in ANTEO+; validity range is also included.

	Unit	Correlation	Validity range [K]
<i>SS316 (Kirillov, 2006)</i>			
c_p	J/(kg K)	$462.0 - 0.134 \cdot T$	300–1700
k_{clad}	W/(m K)	$9.248 + 1.157 \times 10^{-2} \cdot T$	300–1700
<i>15-15 Ti</i>			
c_p	J/(kg K)	$431.0 + 0.177D \cdot T + 8.72 \times 10^{-5} \cdot T^2$	295–1323
k_{clad}	W/(m K)	$8.826 + 1.707 \times 10^{-2} \cdot T - 2.315 \times 10^{-6} \cdot T^2$	293–1273
<i>T91 (Luzzi et al., 2010)</i>			
c_p	J/(kg K)	$-244.0 + 4.677 \cdot T - 1.103 \times 10^{-2} \cdot T^2 + 1.151 \times 10^{-5} \cdot T^3 - 3.979 \times 10^{-9} \cdot T^4$	300–1100
k_{clad}	W/(m K)	$17.94 + 2.51 \times 10^{-2} \cdot T - 1.45 \times 10^{-5} \cdot T^2$	293–894

Table 3

Correlation for gap physical properties implemented in ANTEO+.

	Unit	Correlation	Validity range [K]
<i>Helium (Luscher and Geelhoed, 2011)</i>			
k_{gap}	W/(m K)	$2.639 \times 10^{-3} \cdot T^{0.7085}$	–

**Fig. 4.** Comparison between thermal conductivity correlations for MOX fuel implemented in ANTEO+ at beginning of life.

The last two models are compared in Fig. 4. Correlation of Popov et al. (2000) is plotted without FRA , coherently with the one by Philipponneau (1996) which accounts for fission product effect with a term linearly dependent on burnup and so not present at zero irradiation. For stoichiometric fuels agreement is good even for what concerns the effect of porosity,² while for hypostoichiometric MOX sensible differences are present at low temperatures.

2.8. Constitutive relations for conservation equations

Besides the physical properties, constitutive relations are needed in order to eliminate some of the unknowns present in the conservation equations. In the system given by Eqs. (2), (4), (6), (8) f_i , $\Delta P_{form,i}$, W_{ij}^H and δ must be specified. Moreover, x_i and m_i are available, directly or indirectly, from the boundary conditions; the linear power transferred to a SC is calculated from the power input supplied by the user while the mass flow in a SC is calculated from the flow split model and the total flow of the assembly.

² This is due to the high similarity of the porosity effect in the two correlations.

In this subsection the adopted flow split model and the correlations implemented as constitutive relations are presented, highlighting their expected range of validity.

2.8.1. Friction factor

A great deal of attention was given in the 1970s and 1980s (Rehme, 1973b) to the pressure drops in wire wrapped and bare fuel assemblies, and an important effort was undertaken, in order to establish robust and multipurpose semi-empirical correlations for friction factors. From these experimental results and the ones reported in Chen et al. (2014), four correlations were selected for both bare and wire spaced bundles in both hexagonal and square geometries.

In ANTEO+ the friction factor is actually used twice for two different purposes: the flow split and the bundle average pressure drop (see Eq. (4)). The accuracy and detail requirements can be different, so some model, even if not suited for flow split calculation, can still produce acceptable results in an average sense; nonetheless, selection's consistency is always recommended.

Blasius. The simple correlation proposed by Blasius (1913) for the friction factor in a smooth circular tube reads

$$f_i = \frac{0.316}{Re_i^{0.25}}, \quad (16)$$

where Re_i is the Reynolds number based on the hydraulic diameter of SC i . Blasius does not differentiate among SC types and it is not suited for wire spaced bundles, on the other hand has the advantage of being simple and applicable to a wide range of flow conditions and can thus be used as a first estimate correlation. The range of validity is:

- turbulent $\leq Re \leq 10^5$.

Rehme-bare. As suggested in Rehme and Trippe (1980) the use of the hydraulic diameter equivalence, coupled with friction factors for circular tube can lead to misleading results. To overcome this limitation in Rehme (1973b) a methodology was developed applicable to laminar and turbulent flow regimes and suited for both hexagonal and square geometries.

In the case of fully developed laminar flow the friction factor can be expressed as

$$f = \frac{K}{Re}, \quad (17)$$

where K is a constant depending on SC geometry. The latter is determined by the pitch to diameter ratio P/D (where P is the bundle pitch) for central SCs and, additionally, by the W/D ratio (where W is the maximum pin-duct distance) for edge and corner SCs (see Fig. 1). For every SC family K was computed and plotted over a wide range of P/D and W/D ratios as explained in Rehme and Trippe

(1980). By considering a parallel of all the SCs, the bundle average K_{tot} can also be computed as

$$\frac{1}{K_{tot}} = \sum_i \frac{1}{K_i} \left(\frac{p_{wb}}{p_{wi}} \right)^2 \left(\frac{A_i}{A_b} \right)^3, \quad (18)$$

where p_{wb} is the bundle wetted perimeter.

For turbulent flow a method was developed to predict the friction factor on the basis of laminar results; the following implicit formula was derived

$$\sqrt{\frac{8}{f_b}} = A \left[2.5 \ln \left(Re \cdot \sqrt{\frac{8}{f_b}} \right) + 5.5 \right] - G^*, \quad (19)$$

where A and G^* are empirical factors graphically determined. In order to make the implementation of the model possible in ANTEO+ the interpolating formulae developed by Mattioli and Cevolani (2012) were used, which are very similar to the ones reported in Cheng and Todreas (1986), namely

$$A = \begin{cases} 1.89 - 0.215 \ln(K_{tot}) & \text{if } K_{tot} \leq 64, \\ 1 & \text{if } K_{tot} > 64, \end{cases} \quad (20)$$

$$G^* = 3.3618 + 0.1898 \ln(K_{tot}) + 0.0902 [\ln(K_{tot})]^2 + 0.0007 [\ln(K_{tot})]^3. \quad (21)$$

For dealing with the transition region no indication is reported by Rehme. A possible choice for the extrapolation could be the one used in Cheng and Todreas (1986) (see Eq. (27)), but, in order not to double the approach, a different one has been taken in ANTEO+; specifically, given the Reynolds number, the friction factor is calculated with both the laminar and turbulent formulas, then, the higher is selected as the reference friction factor.

Rehme's correlations for bare bundles can predict friction factors differentiated by SC type or bundle averaged and so it can be used both in the flow split model and in the pressure drops calculation phase.

This model was proved to give reliable results for a broad range of geometric arrangements (Rehme, 1973b) and flow conditions. No explicit validity domain is specified.

Rehme-wire. While the pressure drop due to a spacer grid is linearly added, the wire spaced effect is usually included in the distributed pressure losses. For this reason correlations developed for bare bundles are not transferable to wire spaced ones (although the opposite can be true) for which, in turn, they must be specifically determined. Among the many correlations proposed for wire spaced bundles the one by Rehme (1973a) has been selected and implemented in ANTEO+.

The model is based on an effective velocity v_{eff} created by the swirl flow of the wire, which is a function of P/D and $H/(D + D_w)$, where H is the lead of the wire wraps and D_w is its diameter. The relation is

$$F = \left(\frac{v_{eff}}{v_b} \right)^2 = \sqrt{\frac{P}{D}} + \left[7.6 \frac{D + D_w}{H} \left(\frac{P}{D} \right)^2 \right]^{2.16}, \quad (22)$$

so that the friction factor is finally calculated as

$$f_b = F \left(\frac{p_{w,nw}}{p_{wb}} \right) f^*, \quad (23)$$

where $p_{w,nw}$ is the wetted perimeter of the bundle without the duct and f^* is a modified friction factor expressed as

$$f^* = \frac{64}{Re_b \sqrt{F}} + \frac{0.0816}{(Re_b \sqrt{F})^{0.1333}}. \quad (24)$$

It is declared that this equation is able to predict 92% of around 1400 experimental points within $\pm 5\%$ for $Re \geq 2 \times 10^4$.

This model does not discriminate among SCs and so is less suited for flow split calculations, similarly to Blasius. The validity domain is:

- $1.125 \leq (P/D) \leq 1.417$,
- $8 \leq (H/D) \leq 50$,
- $7 \leq N_{pin} \leq 61$,
- $10^3 \leq Re_b \leq 3 \times 10^5$.

Cheng–Todreas. The correlation which was found to give the best agreement over a very wide range of geometries and flow conditions (Chen et al., 2014; Chun and Seo, 2001) for wire wrapped bundles is the one by Cheng and Todreas (CT) (Cheng and Todreas, 1986). This model is based on theoretical considerations of flow pattern in different types of SCs, enabling different friction factors among SCs to be calculated. From the last point emerges that it can be efficiently cast in a flow split model, moreover, it can be applied to bare or gridded bundles.

A general relationship for f_i can be expressed as (see also Rehme-bare)

$$f_i = \frac{C_{fi}}{Re_i^m}, \quad (25)$$

where m is an index that depends on the flow regime (laminar, turbulent) and C_{fi} , similarly to K , is a function of the SC geometry. The correlation moves from this general form to be detailed for every SC type.³

Interior. It is postulated that the pressure loss originates from two effects:

- 1 the friction loss caused by the fuel rod surface and
- 2 the drag loss caused by the wire.

Under these assumptions the friction factor for central SCs is expressed as

$$f_1 = \frac{C_{f1}}{Re_1^m} = \frac{1}{Re_1^m} \left[C'_{f1} \left(\frac{p'_{w1}}{p_{w1}} \right) + 3W_d \left(\frac{A_{r1}}{A'_1} \right) \left(\frac{D_{H1}}{H} \right) \left(\frac{D_{H1}}{D_w} \right)^m \right], \quad (26)$$

where C'_{f1} is the friction factor constant for a bare rod, p'_{w1} and p_{w1} are respectively the bare and wire spaced rod wetted perimeters, W_d is the wire drag constant, A_{r1} is the projected wire area in the SC, A'_1 is the bare SC area, H and D_w are the wire pitch and diameter respectively. The coefficient n is 1 for laminar and 0.18 for turbulent flow. For a bare bundle A_{r1} is zero and the wetted perimeter ratio is one; the correlation correctly reduces to the bare rod case.

The empirical constants C'_{f1} and W_d are correlated for laminar and turbulent conditions, so that they define two different friction factors named, $f_{1,L}$ in laminar and $f_{1,T}$ in turbulent flow regime. In the transition region f_1 is found using

$$f_{1,tr} = f_{1,L}(1 - \Psi_1^\gamma)(1 - \Psi_1)^\gamma + f_{1,T}\Psi_1^\gamma, \quad (27)$$

where γ and ν are chosen so to best fit the data and Ψ_1 is the intermittency factor.

Edge. Because near the wall the wires sweep the SCs always in the same direction, a swirl flow originates and the coolant is assumed to follow the wire; so only a small velocity component slips over the wire and the drag losses can be assumed negligible

³ In the following the interior SC is denoted with index 1, the edge with 2 and the corner with 3.

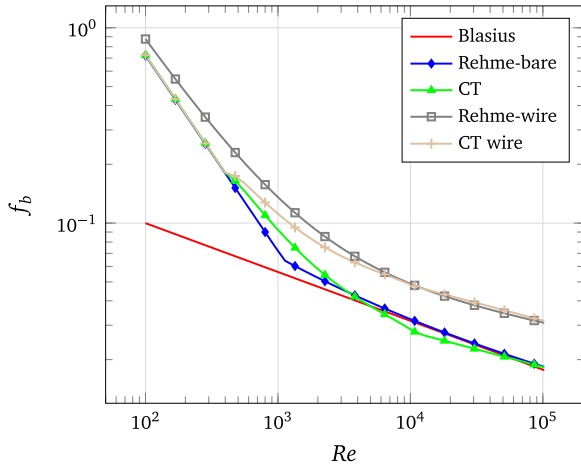


Fig. 5. Comparison between friction factor correlations implemented in ANTEO+ for bare and wire-wrapped bundles. The following parameters were assumed: $(P/D) = 1.1$, $(W/D) = 1.1$ and $(H/D) = 10$.

compared to the skin friction. The friction factor can therefore be expressed as

$$f_2 = \frac{C_{f2}}{Re_2^m} = \frac{C'_{f2}}{Re_2^m} \left[1 + W_s \left(\frac{A_{r2}}{A'_2} \right) \tan^2 \theta \right]^{(3-m)/2}, \quad (28)$$

where W_s is an empirical constant of proportionality, function of geometry and θ is the wrapping angle. The meaning of the other parameters is identical to the one described for the interior SC as is the transition flow regime calculation.

Corner. Since the flow characteristics are similar to the edge SC the same kind of relationship is assumed for f_3 , namely

$$f_3 = \frac{C_{f3}}{Re_3^m} = \frac{C'_{f3}}{Re_3^m} \left[1 + W_s \left(\frac{A_{r3}}{A'_3} \right) \tan^2 \theta \right]^{(3-m)/2}. \quad (29)$$

The validity domain of the CT correlation for all types of SCs is reported to be:

- $1 \leq (P/D) \leq 1.42$,
- $4 \leq (H/D) \leq 52$,
- $19 \leq N_{pin} \leq 217$,
- $50 \leq Re_b \leq 10^6$.

It is claimed that the correlation can predict bundle friction factors with $\pm 14\%$ accuracy for turbulent flow and within $\pm 30\%$ for laminar flow conditions (due to the higher experimental uncertainties in this regime).

In Fig. 5 the correlations are compared (the bare results are for the same bundle but without spacers). In the turbulent region the agreement is good, while the maximum discrepancies are found in the transition region. The Blasius model is highly distant from the others in the laminar region which is outside the correlation domain.

2.8.2. Grid loss coefficient

The concentrated pressure loss for a grid are quite dependent from the spacer design, and no general correlation can be stated applicable to every grid typology. For this reason in ANTEO+ the user can supply in the input his own correlation of the form proposed in Cigarini and Dalle Donne (1988)

$$c_v = \text{MIN} \left(a + b \cdot Re^c + d \cdot Re^e, \frac{f}{\varepsilon g} \right) \quad (30)$$

$$c_s = c_v \cdot \varepsilon g$$

with c_s the grid loss coefficient and ε the area blockage ratio. Pressure drops are then calculated as

$$\Delta P_{grid} = c_s \frac{1}{2} \rho v_b^2, \quad (31)$$

and are linearly added to the other pressure drops contribution (i.e. friction along the bare bundle).

2.8.3. Flow split

Before starting the actual calculation, a way for specifying how the total flow rate distributes in every SC is needed; this task is performed by the flow split model. The main idea behind the model is to follow the hydrodynamic description depicted in Section 2.4 namely, a set of parallel SCs connected only at the inlet and outlet sections; with this picture, and assuming a radially uniform pressure distribution in the inlet and outlet plenum of the pin bundle, the pressure drops of all SCs must be equal. This constraint is translated in the transverse momentum equation (6) or equivalently in its difference form as

$$\Delta P_i = \Delta P_j \quad \forall i, j. \quad (32)$$

The pressure drop is the sum of terms reported in Eq. (4). Assuming an axially constant flow area, neglecting the acceleration pressure term (first term on the right hand side) and eliminating the hydrostatic pressure, which is equal for every SC, it is possible to recast (32) in the form

$$v_i^2 \frac{f_i}{D_{Hi}} L + \xi_i v_i^2 = v_j^2 \frac{f_j}{D_{Hj}} L + \xi_j v_j^2 \quad \forall i, j, \quad (33)$$

where L is the bundle length and ξ_i is the concentrated loss coefficient of the grid spacers which is equal to the product of the spacers number and c_s described in Section 2.8.2. The form pressure drop term has been related to the dynamic pressure and the term $\frac{1}{2} \rho$ has been simplified, neglecting density variations between SCs.⁴ The flow split parameters are defined as the ratio between the SCs velocity and the average bundle velocity, so Eq. (33) can be written as the ratio of two flow split parameters as

$$\frac{X_i}{X_j} = \sqrt{\frac{f_j(X_j)(L/D_{Hj}) + \xi_j}{f_i(X_i)(L/D_{Hi}) + \xi_i}} \quad \forall i, j, \quad (34)$$

where the dependency of the friction factor on Reynolds number and thus on the velocity and flow split parameters is explicitly reported. It is seen that equation (34) is non-linear.

The system of equations composing the flow split model is closed with the continuity condition

$$\sum_{i=1}^{N_{typ}} S_i X_i = 1 \quad \text{where} \quad S_i = \frac{N_i A_i}{A_b}, \quad (35)$$

where N_{typ} is the total number of SCs types (Figs. 1 and 11) in the bundle, S_i is the fractional area occupied by SCs of type i , N_i is the number of SCs of the i th type and A_b is the bundle flow area. Given the generally non-linear character of the model⁵ the method of successive approximations is used.

Various flow split models can be constructed from the general one just described, depending on the correlation used for the friction factors. Some models account for the difference in friction among SC types (e.g. CT) while others use the same f_i formula for all the SCs (e.g. Blasius) and are essentially based on the hydraulic diameter concept. The former type is generally more accurate, but the latter can still produce reliable results if all the f_i are close to each

⁴ This is consistent with the forced flow assumption.

⁵ If the concentrated loss coefficient is zero the equation can be cast in an explicit form which does not need non-linear iterations.

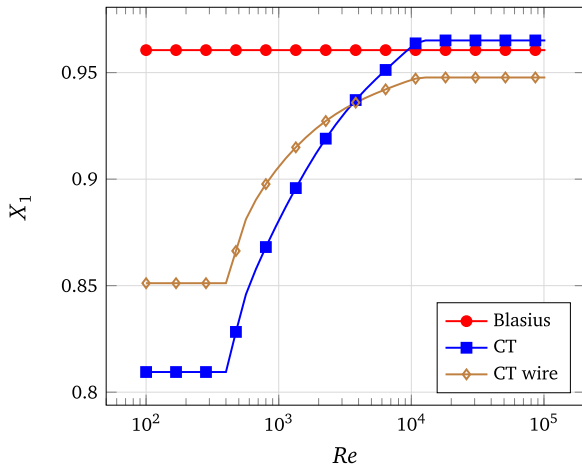


Fig. 6. Comparison between flow split models implemented in ANTEO+ for bare and wire-wrapped bundles. The following parameters were assumed: $(P/D)=1.1$, $(W/D)=1.1$ and $(H/D)=10$.

other; this is partially true for turbulent flow conditions, where geometric characteristics become less and less important (see Fig. 6). In laminar flow, however, not considering the difference among SCs can produce unacceptably inaccurate results. With wire spacer the situation worsens. For these reasons it is always recommended to rely on models which differentiates f_i by SC type (Pramuditya and Takahashi, 2013).

2.8.4. Mixing coefficient

Temperature prediction is probably the most important task of thermal-hydraulic design and so a high degree of accuracy must be targeted in a wide range of situations compatibly with the anticipated range of application of ANTEO+. For this, correlations for W_{ij}^H encompassing both bare and wire spaced bundles have been implemented. The additional mixing effect of the grid spacer is not directly modeled, but it is partially accounted for in the flow split model.

The reference correlation implemented in ANTEO+ for bare or grid spaced bundles, taken from Kim and Chung (2001), is firstly presented.

Kim. In the model proposed by Kim and Chung (2001) the mixing rate is estimated on the basis of a scale analysis on the flow pulsation generated by periodic vortices, that is indicated to be the main cause of mixing in rod bundles. Based upon the assumption that turbulent mixing is the sum of molecular motion, isotropic turbulent motion and flow pulsations (anisotropic turbulence due to heterogeneity effects), the scale relation is derived as a function of P/D , Re , and the Prandtl number Pr .

The effective mass exchange rate between SCs for energy transfer per unit length can be expressed as

$$W_{ij}^H = \rho_{ij} v_{eff} S = \rho_{ij} v_{ij} St_g S, \quad (36)$$

where v_{eff} is the effective mixing velocity, possibly different from v^* , v_{ij} is the mean axial speed in SCs i and j , and St_g is the gap Stanton number which is expressed as

$$St_g = \frac{v_{eff}}{v_{ij}} = \frac{2}{\gamma_1^2} \sqrt{\frac{\alpha}{8}} \frac{D_{Hij}}{s} \times \left[\left(\frac{\gamma_1^2}{2PrRe^{(1-\frac{\beta}{2})}} \frac{8}{\alpha} + \frac{1}{Pr_T} \right) \frac{s}{b\eta} + a_x \frac{z_{FP}}{D} Str \right] Re^{-\frac{\beta}{2}}, \quad (37)$$

where α , β and γ_1 are empirical constants, b is as shape factor equal to 1 in square and 2/3 in hexagonal geometry, D_{Hij} is the mean

hydraulic diameter of SCs i and j , Pr_T is the turbulent Prandtl's number relating eddy diffusivity of momentum and energy, Str is the Strouhal's number, calculated with the formula found in Wu and Trupp (1994), and z_{FP}/D is the path length to diameter ratio.

The correlation already includes the effect of conduction and so the last term in Eq. (8) is automatically included in W_{ij}^H . This correlation can be applied to both square and hexagonal pins' arrangements and was found in good agreement with the published experimental results with various Prandtl number fluid flows, especially for liquid metal coolants ($Pr \ll 1$).

Wire wrapping spacers have been extensively used in assembly design of LMFR (mostly sodium) (Waltar et al., 2013) because, besides holding the fuel pin in position, they enhance mixing mechanisms, although increasing pressure drops over the core. The presence of the helical wire generates a periodic transverse flow pulsation, with frequency inversely proportional to the wire lead pitch. In the interior SCs the wire sweeps the flow in all directions, while in the peripheral SCs it passes in only one direction thus originating a swirl flow (i.e. the flow follows the wire). This pulsation creates pressure waves which are responsible for the enhanced pressure drops and mixing.

The flow pulsation is a local effect since it depends on the position of the wire relative to the gap: when the wire passes the gap the actual mass and energy exchange takes place. ANTEO+, however, does not deal with local correlations or phenomena but with quantities averaged over a representative length (i.e. the wire lead pitch), for this reason the position of the wire will be ignored. To be sure that this average is truly representative the lead pitch must be shorter than the bundle axial height.

Three correlations have been implemented in the code with various degree of modeling and empiricism, in order to give the user a broad spectrum of choices, as was done for the friction factor.

Nijsing. The correlation proposed in Nijsing and Eifler (1974) is based only on theoretical geometric considerations and it does not involve any kind of data fitting; for this reason it is probably the simplest of all the wire mixing models.

The main idea behind this method is that the wire sweeps an annular sector as it passes through the gap between channels; this, considering that the mixing is a periodic event with period proportional to half the wire lead pitch, suggests that the average mixing rate per unit length can be written as

$$W_{ij,wire}^H = \frac{1}{2} D_w \pi \frac{D + D_w}{H} \rho_{ij} v_{ij}, \quad (38)$$

where $W_{ij,wire}^H$ is the mass exchange for energy transfer solely due to the wire; to complete the range of mixing phenomena, pure turbulent and conduction effects must be linearly added. These effects are taken into account in the model by Kim previously highlighted that can thus be used for this superposition.

The above relation is valid for internal SCs only; for edge and corner SCs, the sweep is unidirectional so that the expression becomes

$$W_{ij,wire}^H = D_w \pi \frac{D + D_w}{H} \rho_{ij} v_{ij}. \quad (39)$$

Neglecting differences in velocity, the wire effect is, in the wall region, twice the one in the internal zone. Because no data fitting was performed, no specific validity range is prescribed.

Zhukov. Contrary to the correlation by Nijsing this one is mainly experimental. The founding theoretical consideration of the model is the periodic nature of the sweeping effect assumed to be represented only by the first harmonic (Bogoslovskaya et al., 2000). The mixing rate is expressed as

$$W_{ij,wire}^H = \rho_{ij} v_{ij} \frac{A_i + A_j}{2} \frac{1.047\sigma}{H} \Phi \left(\frac{P}{D} \right) \Psi(Re) \sin(\varphi_{ij}), \quad (40)$$

where $\varphi_{ij} = (2\pi z/H) - \alpha_{ij}$ and α_{ij} is the wire wrap phase entering the j th SC from the i th one. Averaging the last expression over the wire lead pitch we finally arrive at

$$W_{ij,wire}^H = \rho_{ij} v_{ij} \frac{A_i + A_j}{2} \frac{\sigma}{3H} \Phi\left(\frac{P}{D}\right) \Psi(Re), \quad (41)$$

where σ is a parameter relating efficiencies in mass and energy transfer assumed equal to 0.7, Φ is an empirical function of the pitch-to-diameter ratio and Ψ is an empirical function of the Reynolds number. A similar approach is used for wall SCs. This model, similarly to the one by Nijssing, calculates only the wire contribution to thermal mixing and must thus be coupled to the model by Kim.

The formula accuracy is claimed to be 10% inside the validity domain:

- $1.01 \leq (P/D) \leq 1.4$,
- $2 \leq (H/D) \leq 50$,
- $2000 \leq Re_b \leq 2 \times 10^5$.

Cheng–Todreas. Consistently with the friction factor another correlation of Cheng and Todreas (CTm) (Cheng and Todreas, 1986) has been implemented. Contrarily to other correlations this one is of the comprehensive type since it encompasses all principal types of mixing and not just the flow sweeping of the wire (see also Section 2.8.5). The flow sweeping effect is handled in different ways for interior and wall SCs; in fact in the internal ones the wire sweeps flow in a bidirectional fashion similar to what happens with turbulent exchange, and so these two mechanisms are grouped together, while for wall SCs the sweep effect is unidirectional and so is treated separately from other terms (i.e. linearly added).

Interior. In the interior SCs the wire sweeping and the turbulence effects are correlated together by means of the eddy diffusivity ϵ_{ij} concept, which is linked with W_{ij}^H by

$$W_{ij}^H = \frac{\rho_{ij} \epsilon_{ij} S}{\eta}. \quad (42)$$

In the actual correlation the dimensionless parameter $\epsilon_{ij}/v_{ij}\eta$, similar to the Stanton's number, is used.

Wall. For wall SCs (i.e. edge and corner) the unidirectional flow sweeping effect is expressed as

$$W_{ij,wire}^H = s \rho_{ij} v_{ij} C_1, \quad (43)$$

where C_1 is the dimensionless transverse velocity defined as the ratio between the transverse and axial velocity components. In the last equation it was supposed that the transverse velocity is the same for all wall SCs (this assumption stems from considerations on the continuity equation and the constant mass flow assumption).

$\epsilon_{ij}/(v_{ij}\eta)$ and C_1 are correlated for both laminar and turbulent conditions; in the transition region, the intermittency function is used similarly as for the friction factors.

The proposed validity range is:

- $1.067 \leq (P/D) \leq 1.35$,
- $4 \leq (H/D) \leq 52$,
- $7 \leq N_{pin} \leq 217$,
- $400 \leq Re_b \leq 10^6$,

in which $\epsilon_{ij}/(v_{ij}\eta)$ and C_1 can be predicted with 25% and 15% accuracy respectively.

The three correlations for wire spaced bundles are compared in Fig. 7. The simple approach proposed by Nijssing is not in agreement with the other experimental correlations for this geometric arrangement.

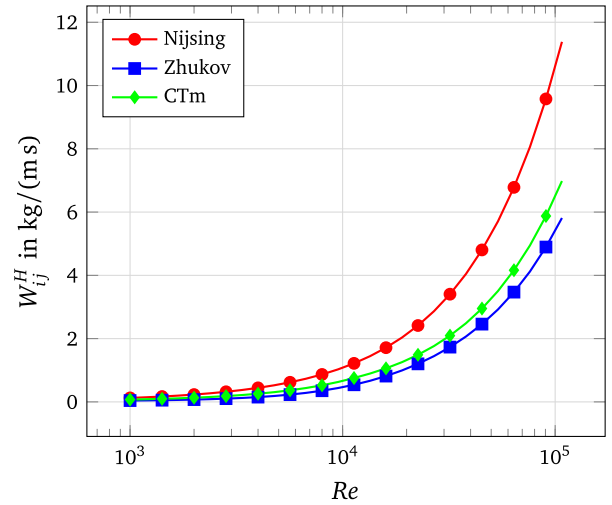


Fig. 7. Comparison between wire thermal mixing correlations implemented in ANTEO+. The following parameters were assumed: $(P/D) = 1.1$, $(W/D) = 1.1$, $(H/D) = 10$ and $Pr = 0.025$ corresponding to lead. The contribution given by Kim's correlation has been added to Zhukov and Nijssing.

2.8.5. Conduction shape factor

The conduction shape factor κ appears in the conduction term in the energy equation (8). In this, the temperature gradient is approximated as a finite difference and the representative length is taken as the centroid-to-centroid distance η which is not, however, the correct one. The conduction shape factor is introduced in order to correct the finite difference and stems from the lumped nature of the SC approach.

In ANTEO+ two correlations have been implemented.

Cheng–Todreas. As stated in Section 2.8.4 the mixing correlation of Cheng and Todreas (Cheng, 1984) is comprehensive, and as such even molecular heat conduction is modeled. In particular κ is expressed as

$$\kappa = 0.66 \left(\frac{P}{D}\right) \left(\frac{s}{D}\right)^{(-0.3)}, \quad (44)$$

which was proposed for bare bundles and is assumed to be applicable also to wire spaced ones. It has to be mentioned that it can only be applied to hexagonal arrangements.

Ma. In Ma et al. (2012) a method was developed to make use of the correlations and experimental data of annuli to obtain the heat transfer rate of LMCs flowing in rod bundles. Their interaction correction factor was found to be generally related to the conduction shape factor and correlated as

$$\kappa = 0.5 \left(\frac{\frac{D_{an}}{D} - 1}{\frac{P}{D} - 1} \right), \quad (45)$$

where D_{an} is the outer diameter of the equivalent annulus. The correlation can be applied to square and hexagonal bundles.

The two correlations are compared in Fig. 8 for hexagonal geometry. They are in good agreement in the plotted range.

2.9. Nusselt number

For calculating the cladding outer temperature with Eq. (9) the convection coefficient α_p must be determined; this can be done via the Nusselt's number.

In ANTEO+ two correlations have been implemented: the first is the one by Mikityuk (2009) for hexagonal bundles while the second was conceived by Ma et al. (2012) in parallel with the conduction shape factor described in Section 2.8.5. The latter is applicable to both square and hexagonal pin arrangements.

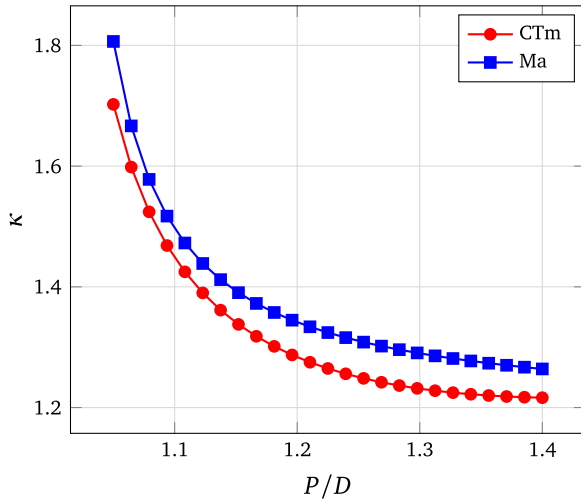


Fig. 8. Comparison between correlations for the conduction shape factor implemented in ANTEO+.

2.9.1. Mikityuk

The correlation for Nu has the form

$$Nu = 4.7 \times 10^{-2} \left(1 - \exp \left(-3.8 \left(\frac{P}{D} - 1 \right) \right) \right) (250 + (Re \cdot Pr)^{0.77}). \quad (46)$$

Recently in [Pacio et al. \(2014\)](#) the authors compared this correlation with experimental data for lead–bismuth in hexagonal geometry; satisfactory agreement was found.

2.9.2. Ma

The correlation has the following form,

$$Nu = \left(4.82 + 0.697 \cdot y + 0.022 \left(\frac{Re \cdot Pr}{Pr_T} \right)^{0.758 \cdot y^{0.053}} \right) C_h C_t C_i C_s, \quad (47)$$

where y is the ratio between the external and internal radius of the equivalent annulus while C_h , C_t , C_i and C_s are correction parameters accounting for the annulus approximation.

The correlations are compared in [Fig. 9](#) where we see the good agreement at low P/D which decreases at higher values. Moreover

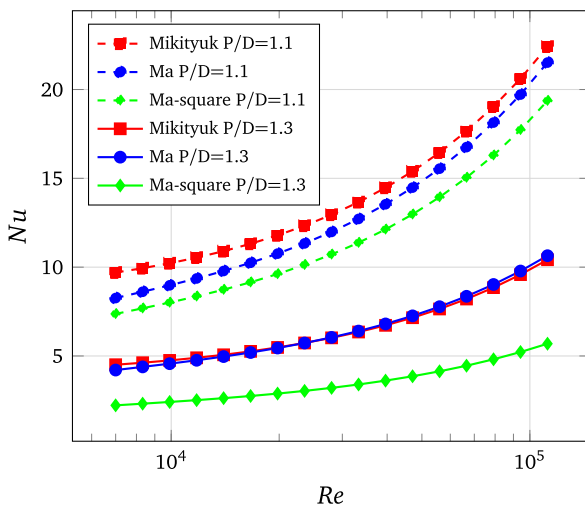


Fig. 9. Comparison between Nusselt's number correlations implemented in ANTEO+. The Prandtl number has been set to 0.025 corresponding to lead. Ma-square corresponds to the correlation by [Ma et al. \(2012\)](#) for square bundles.

we observe that at higher P/D the difference between the results for hexagonal and square bundles diminishes.

2.10. Bypass model

Usually thermal hydraulic calculations at SA level are performed with the hypothesis that the bundle is adiabatic and so no heat is lost to the wrapper or other SAs. This holds true if thermal gradients between SAs are small, as near the core center (in standard configurations, e.g. without internal blankets). If a SA is placed side-by-side with a dummy element, a control or safety rod assembly, or if we are near the periphery this could not hold true. Assuming that a pins bundle is adiabatic is however a conservative assumption, especially for the hottest one, so for safety reasons is acceptable. The above procedure has one disadvantage, namely, that the wrapper temperature itself cannot be estimated with some accuracy; moreover the effect the bypass has on the outer rows of pins is not accounted for. It can be important to know such details in the verification phase to be sure that the thermal gradients in the duct are not high enough to imply such a thermal expansion that will worsen heat transfer of adjacent pins. In order to calculate such effects a simple model has been devised and implemented in ANTEO+. The model is divided into two parts: first comes the calculation of the bypass flow rate if requested, otherwise a selected flow rate can be used; second is the calculation of the temperature distribution of the assembly coupled to the wall and bypass regions. The following sections briefly describes model's features and range of applicability.

2.10.1. Bypass flow rate

A flow split model conceptually identical to the one presented in [Section 2.8.3](#) has been implemented where distributed and concentrated friction coefficients must be supplied, along with geometrical details of the flow path of the bypass and bundle regions.

2.10.2. Heat transfer

By performing a scale analysis, it can be shown that the ratio between the bundle outward Q_{out} and the transverse (i.e. perpendicular to the outward direction) Q_{tr} heat flux can be approximated, in typical fast reactor operating conditions, as

$$\frac{Q_{out}}{Q_{tr}} = \frac{\Delta T_w}{T_{sc} - T_{bp}} \left(\frac{s_w}{P} \right)^2, \quad (48)$$

where ΔT_w is a temperature difference between two wrapper nodes, T_{sc} and T_{bp} are respectively a representative SC and bypass temperatures and s_w is the wrapper thickness. The above ratio measures how much two nodes are thermally similar: if it is low the nodes could be ideally grouped together without much loss of accuracy, but if it is high the nodes must be treated separately, accounting for their coupling. It is usually small,⁶ but in presence of strong power gradients or uneven cooling of corner and edge SCs, however, the flux ratio can increase. For properly modeling these phenomena the most logical subdivision for the wrapper and bypass region is the same one used in the SC definition (see [Fig. 10](#)).

The energy conservation equation presented in [Section 2.4](#) must be augmented with the exchange term between SCs and wrapper, which has the form

$$\frac{T_w - T_i}{R_t}, \quad (49)$$

⁶ A similar analysis leads to the same conclusions for the transverse heat flux in the bypass region.

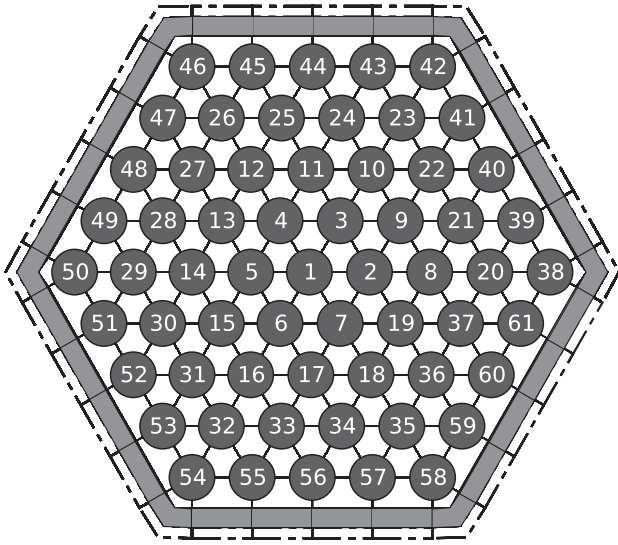


Fig. 10. Representation of the nodalization scheme used in ANTEO+ for performing calculations including the bypass region between assemblies.

where R_t is the thermal resistance between the SC and wrapper node centers, which is written as

$$R_t = \frac{1}{\alpha_{wr} A_{wr}} + \frac{s_w/2}{k_w A_{wr}}, \quad (50)$$

where α_{wr} is the convection coefficient for thermal exchange between SC and wrapper, A_{wr} is the contact surface for the exchange and k_w is the wrapper thermal conductivity. The selected Nu_{wr} correlation used to calculate α_{wr} is the one presented by [Sleicher and Rouse \(1975\)](#) for pipes flow, which was proven in [Kawamura et al. \(1998\)](#) to have the closest agreement with DNS (direct numerical simulations) calculations for a uniformly heated rectangular duct. This correlation is also recommended in [OECD/NEA \(2007\)](#) and has the following form

$$Nu_{wr} = 5 + 0.015 \cdot Re^a \cdot Pr^b, \quad (51)$$

$$a = 0.88 - \frac{0.24}{4 + Pr},$$

$$b = \frac{1}{3} + 0.5 \exp(-0.6Pr).$$

The reported validity range is:

- $Pr \leq 10$,
- $10^4 \leq Re \leq 10^6$.

The energy balance of the bypass region is identical to the one outlined for the bundle SCs (as is the Nu correlation), but without the power discharged by the pin. A term for simulating power deposition in this region has also been added.

The conservation statement for a generic node in the duct region can be written as

$$\frac{T_{sc} - T_w}{R_t} = q_{wr} + \frac{T_w - T_{bp}}{R_t} + \sum_j \Delta z \frac{s_w k_w}{\delta_w} (T_w - T_{w,j}), \quad (52)$$

where q_{wr} is the power deposited in the wrapper and δ_w is the distance between wrappers' nodes. The R_t factors in the right and

left hand sides of Eq. (52) are different only for a geometrical effect, solely present in hexagonal bundles.

2.11. Geometric capabilities

ANTEO+ has extended geometric capabilities which encompass both hexagonal and square arrangements enclosed in a casing duct. The possibility of simulating pins of different size, structural rods, guide tubes with or without control rods and edge channel displacer has been implemented into the code in order to give the user some modeling flexibility. Internal structural walls, like in the ELSY (European Lead-cooled SYstem) design ([Artioli et al., 2009](#)), are also available both in hexagonal and square geometries. Finally the innovative design feature proposed in [Tobita et al. \(2013\)](#) is also present in ANTEO+; this consist of an inner discharge duct which prevents coherent motion of the melted fuel within the bundle lattice and so reducing risk of recriticality, which is the main issue in case of CDA (Core Disruptive Accident). The above geometrical features are schematically illustrated in Fig. 11.

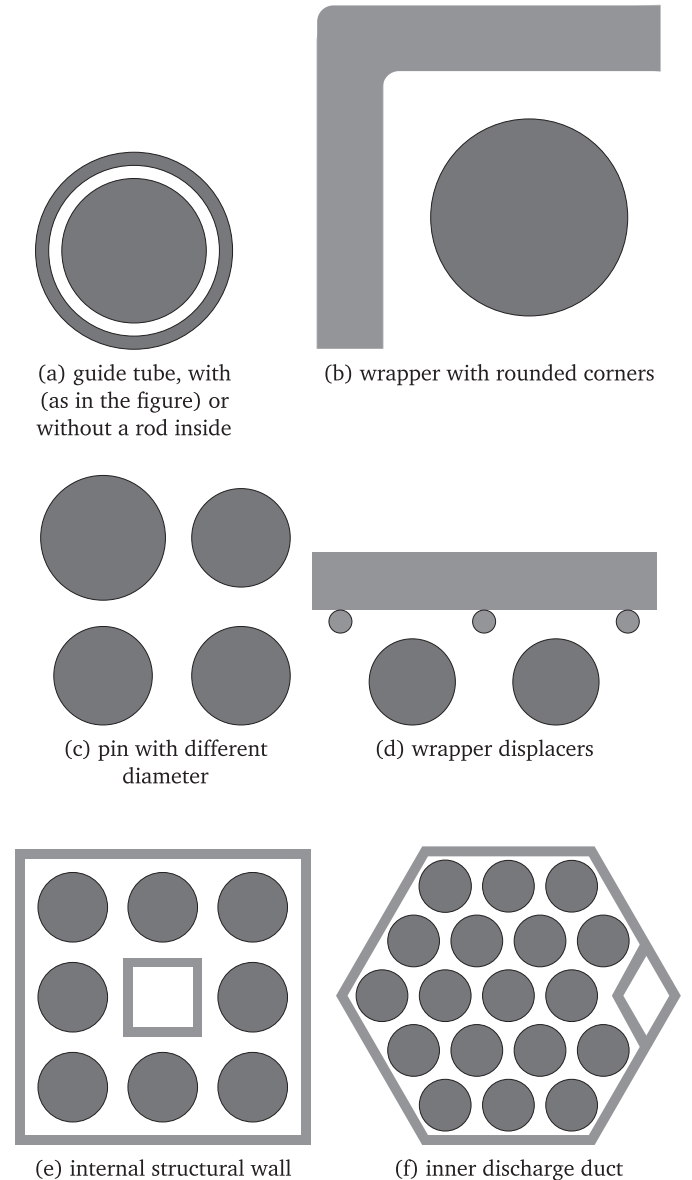


Fig. 11. Examples of ANTEO+ geometrical modeling capabilities.

Table 4
Collection of experiments used for flow split models validation.

Ref.	N_{pin}	P/D	H/D	$Re(\times 10^3)$
<i>Laminar</i>				
Cheng (1984)	61	1.250	24.0	0.64
Cheng (1984)	61	1.250	48.0	0.64
Cheng (1984)	19	1.245	35.2	0.32
<i>Turbulent</i>				
Cheng (1984)	217	1.283	48.0	20.0
Cheng (1984)	61	1.250	24.0	4.50
Cheng (1984)	61	1.250	48.0	4.50
Cheng (1984)	91	1.210	48.0	20.0
Cheng (1984)	37	1.190	34.8	14.0
Cheng (1984)	217	1.257	51.7	73.0
Cheng (1984)	91	1.240	48.0	20.0
Miyaguchi (1979)	169	1.210	47.1	32.0–58.0
Rehme and Trippe (1980) ^a	19	1.300	bare	60.0
<i>Complete range</i>				
Chiu (1977)	37	1.155	21.0	3.00–13.9
Cheng (1984)	61	1.067	4.0	2.00–14.6
Cheng (1984)	61	1.067	8.0	1.00–21.8
Cheng (1984)	217	1.250	51.7	1.80–12.7
Cheng (1984)	37	1.154	13.4	3.47–26.3

^a Only data set for gridded bundle.

3. Validation

To ultimately estimate, in a quantitative way, the degree of reliability and range of application of a code, the comparison with experimental data, representative of the operating conditions and geometric configurations to which the code will be applied, is a mandatory step. This is part of the validation phase of a code.

SC codes, in particular, have empirical basis and so understanding and clearly identifying their application domain can heavily help the user in the results interpretation phase. For this, in ANTEO+, the application range was decided beforehand (see Section 1) and thus validation takes on a slightly different meaning, namely, to verify that the anticipated and actual validity domains coincide, thus supporting the derived set of equations and modeling approximations. As explained in Section 2 the expected range of application for ANTEO+ is the single phase forced convection regime, and consequently this is the main focus of the present work. Nonetheless, experiments falling out of this regime are analyzed in order to quantify extrapolation errors.

In the present section, the models previously presented, encompassing flow split, pressure drops, SC temperatures and Nusselt number are thoroughly validated; then, in the subsequent section results are critically discussed.

To better contextualize ANTEO+ accuracy, a quantitative comparison with other codes, like COBRA-IV-I-MIT (Fricano, 2012) and ENERGY-II (Cheng, 1984), is reported. This will also highlight the advantages of ANTEO+, within its validity domain, over the historical one outlined in Section 1.

3.1. Flow split

As explained in Section 2.8.3 the flow split model could be coupled with any one of the friction factors described in Section 2.8.1. In ANTEO+ however the flow split model can be used only in combination with the Blasius or CT friction factor models; the former does not account for the different friction factors in each SC type and so can only be used for bare or gridded bundles in the turbulent region. The latter is more general encompassing wired, gridded and bare bundles in every flow regime.

The experimental data used in the validation are listed in Table 4. The data set is basically the one used in Cheng (1984) with the addition of three cases. The results are summarized in Table 5 and

Table 5
Summary of flow split models validation. Results are presented differentiating flow regimes.

	$\bar{\varepsilon}$ [%] ^a	σ [%] ^b	RMS [%] ^c
<i>Total (117 points)</i>			
CT	4.8	6.8	6.8
Blasius	9.8	19.3	13.2
<i>Turbulent region (43 points)</i>			
CT	1.8	2.1	2.2
Blasius	5.9	9.7	9.6
<i>Turbulent gridded bundle (3 point)^d</i>			
CT	1.2	–	1.2
Blasius	3.4	–	3.4

^a Defined as $(\sum_{i=1}^n |x_{exp} - x_{calc}|)/n$.

^b Standard deviation.

^c Root mean square.

^d No standard deviation is reported due to insufficient statistic.

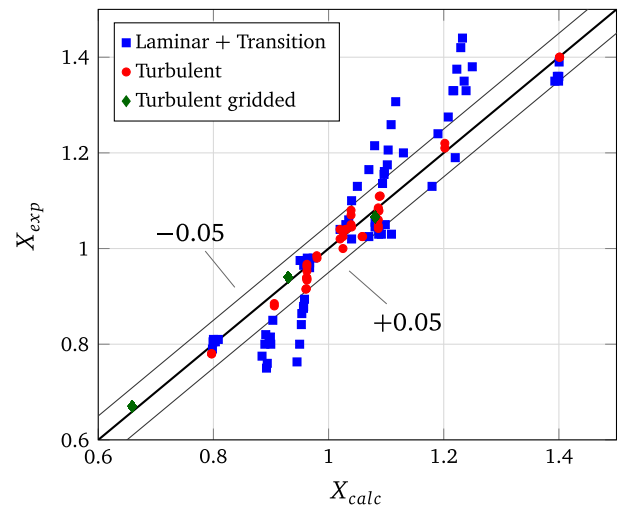


Fig. 12. Comparison of CT flow split model with experimental data used in the validation.

Figs. 12 and 13; in these, three cases are reported: the whole ensemble of data point analyzed, the data in the turbulent region only and the data for bare bundles in the turbulent region. The last category is chosen according to the validity range of the Blasius based flow split model.

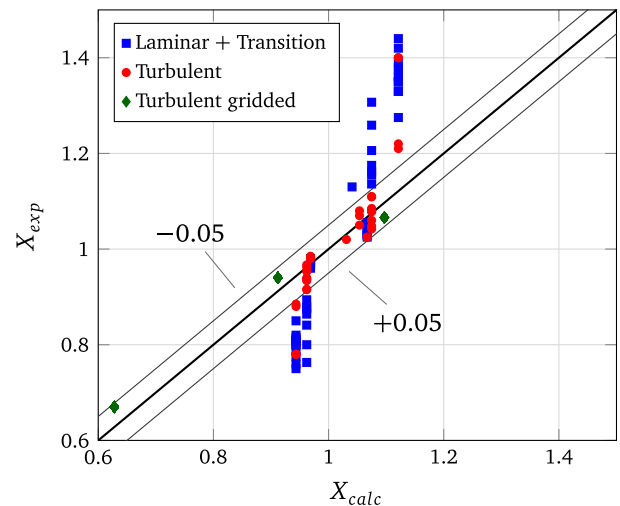


Fig. 13. Comparison of Blasius flow split model with experimental data used in the validation.

Table 6

Collection of experiments used for pressure drops validation. Experiments cover only grid spaced bundles.

Ref.	Liquid	N_{pin}	P/D	$Re(\times 10^3)$
Schikorr et al. (2010)	Water	19	1.40	8.0–73.0
Litfin et al. (2011)	Lead–Bismuth	19	1.40	21.1–55.5

Table 7

Results summary for pressure drops validation of grid spaced bundles.

	$\bar{\epsilon}_{rel} [\%]^a$	$\sigma [\%]$	RMS [%]
Total (21 points)			
CT	3.3	4.1	3.9
Rehme	6.5	8.0	8.1
Blasius	4.6	5.8	5.7

^a Defined as $(\sum_{i=1}^n \frac{|x_{exp} - x_{calc}|}{x_{exp}}) / n$.

3.2. Pressure drops

The bundle pressure drop ΔP_b can be expressed as

$$\Delta P_b = f_b \frac{1}{2} \rho v_b^2 \frac{L}{D_H}, \quad (53)$$

where D_H is the bundle hydraulic diameter. From Eq. (53) we see that, neglecting geometric uncertainties, the validation of the friction factor and pressure drops are equivalent, and, as such, the latter is used for validating the friction factor models presented in Section 2.8.1.

The friction factor must be validated for bare, grid spaced and wire spaced bundles; concerning the latter, in a recent article (Chen et al., 2014), the authors compared a number of correlations with a broad set of experimental data. In their analysis the two correlations for wired bundles presented in Section 2.8.1, namely CT and Rehme-wire, were among the correlations analyzed. The authors concluded that the CT correlation has the best performances while Rehme-wire ranked lower. Because we deem their analysis representative of the anticipated application range of ANTEO+, the validation of the friction factor and pressure drops for wire spaced bundles is not reported here, but borrowed from Chen et al. (2014).

The present section is therefore focused on the validation of gridded bundles pressure drops; experimental grid pressure losses, however, are subtracted from the overall ones in order to retrieve bare bundle results which are, in turn, directly comparable to the correlation reported in Section 2.8.1. This choice stems from the wish of keeping the two effects separated and from the already proven ability of the general form of the grid loss coefficient presented in Section 2.8.2 of reproducing experimental data (Epiney et al., 2010).

The experiments used in the validation are reported in Table 6. One, among the selected, employs water as a coolant, but because it is an isothermal zero power experiment, it is still a representative case for validating pressure drops of LMCs. A summary of the results can be found in Table 7 and Figs. 14–16.

3.3. Temperatures

The assessment of the ability of a SC code in reproducing experimental steady state SCs temperature distributions is of paramount importance. A SC temperature can effectively be a tight design constraint, moreover it is used in estimating the thermal-hydraulic feedback effects on neutronic performances and as initial condition in safety analyses. Finally, in the evaluation of SC temperatures distribution, the overall performances of the code can be effectively tested, because, in solving the conservation equations all the models outlined in Section 2.8 play along with the inherent approximations of ANTEO+. For these reasons a thorough validation is now presented.

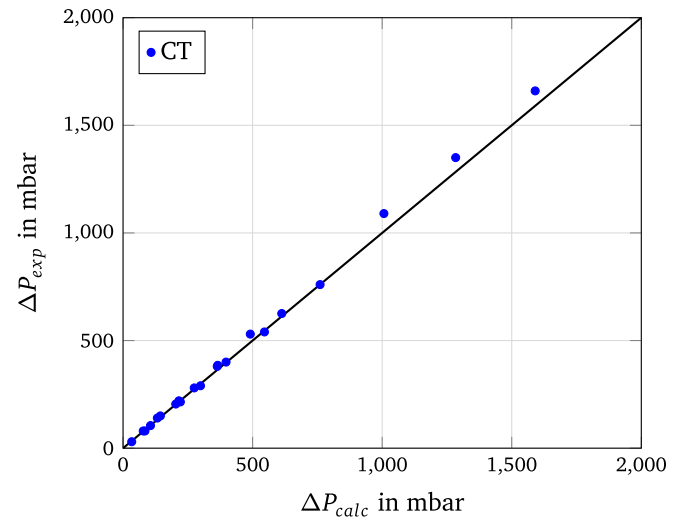


Fig. 14. Comparison of CT model with experimental data used in the validation of pressure drops for grid spaced bundles.

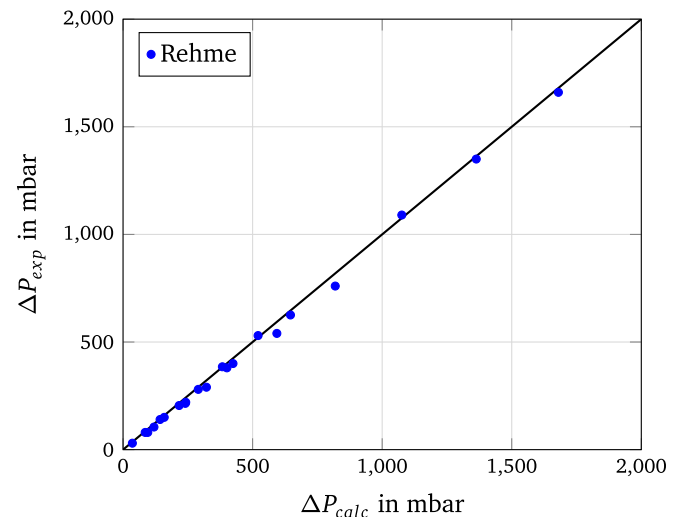


Fig. 15. Comparison of Rehme model with experimental data used in the validation of pressure drops for grid spaced bundles.

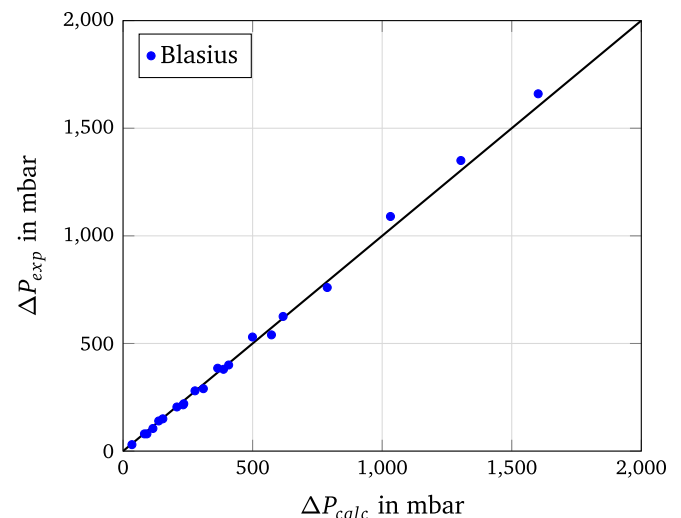


Fig. 16. Comparison of Blasius model with experimental data used in the validation of pressure drops for grid spaced bundles.

Table 8

Comparison between calculated and experimental bundle average temperature gain for the experiment described in Litfin et al. (2011).

	$\bar{\epsilon}_{rel}$ [%]	σ [%]	RMS [%]
Total (24 points)	1.1	1.3	1.5

Table 9

Collection of experiments used for SC temperature validation. Main physical and geometrical parameters are also given.

Ref.	Coolant	N_{pin}	P/D	H/D	$Re(\times 10^3)$
<i>Hexagonal – wire spacer</i>					
Fontana et al. (1973)	Na	19	1.243	52.2	0.90–68.9
Namekawa et al. (1984)	Na	37	1.210	47.2	0.86–11.9
Morris et al. (1980)	Na	61	1.243	52.2	49.0–73.0
Engel et al. (1980)	Na	61	1.082	7.70	0.50–13.0
Rameau and Menant (1984)	Na	91	1.067	11.4	0.60–3.27
<i>Square – grid spacer</i>					
Zhukov et al. (2004)	NaK	25	Variable ^a	–	53.4

^a Pins with different diameters are present.

3.3.1. Bundle temperature

Before starting with the local SC assessment an integral experiment is analyzed, in order to ascertain the accuracy of some of the equations of state presented in Table 1. The experiment taken from Litfin et al. (2011), already used in Section 3.2, employs lead–bismuth as a coolant. The results are summarized in Table 8 and Fig. 17.

3.3.2. SC temperatures

A sizeable number of cases has been simulated as reported in Table 9, trying to cover the anticipated application domain of ANTEO+. As stated at the beginning of Section 3, cases outside the anticipated validity range of ANTEO+, namely, single phase forced convection, have been also simulated; these cases fall in the so called mixed convection regime in which natural and forced convections are equally important. Mixed convection cases are only used, in the present work, to highlight in a more quantitative way extrapolation errors and make the reader more aware of the consequences of violating the validity range of ANTEO+. For a discussion on mixed convection modeling problems and specificities the reader is referred to Iannello et al. (1988), Suh et al. (1989a,b) and Suh and Todreas (1987). In order to quantify the degree of natural convection and thus how much the operating conditions are

Table 10

Comparison summary between simulations and ORNL-19 bundle experimental data.

Identifier	Power (kW)	Flow rate (kg/s)	Y	$\bar{\epsilon}$ [%]	
				ANTEO+	COBRA
Power Uniform					
Test 2 - Run 105	322.3	3.06	0.09	3.9	5.4
Power skew 1.25					
Test 4 - Run 105	31.13	0.27	0.73	3.7	-
Test 4 - Run 101	4.940	0.07	1.00	0.9	0.9
Power skew 3.0					
Test 14 - Run 101	16.52	2.99	0.05	5.4	-

^a Skew defined as the maximum-to-minimum power ratio.

far from ANTEO+ domain, a parameter called Y taken from Engel et al. (1980) has been used; Y is defined as

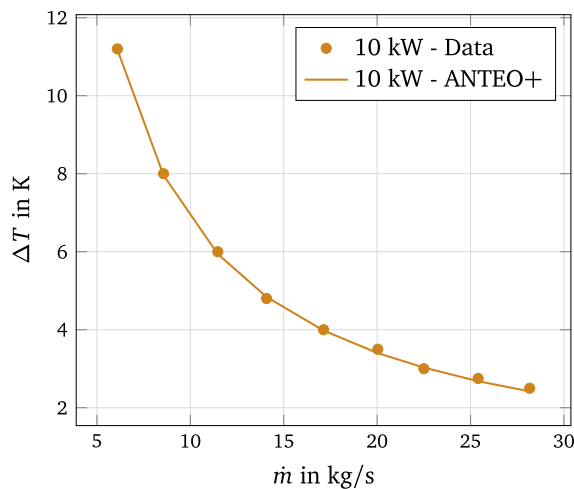
$$Y = \frac{2Gr}{fRe^2}, \quad (54)$$

where Gr is the Grashof's number based on the temperature film drop. Y is basically the ratio of the transverse buoyancy forces, drivers of radial thermal mixing and axial pressure drops, responsible for axial mass flow.

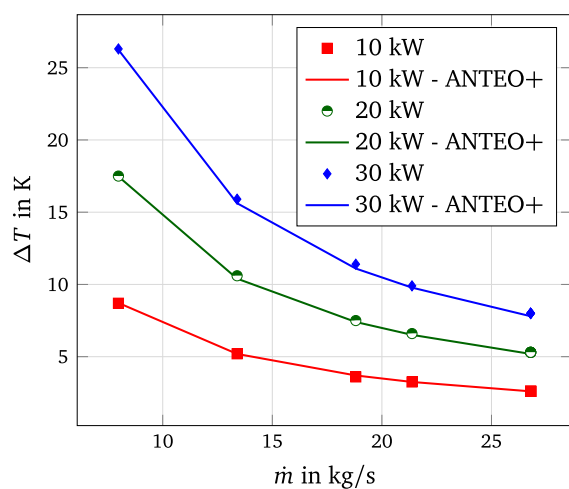
From the results outlined in Engel et al. (1980) the value of 1.5×10^{-3} has been detected as the discriminant between forced and mixed convection; this corresponds to a reduction of approximately 10% of the maximum temperature gain relative to fully forced convection. Henceforth the reported values of Y will be normalized to the reference value; if greater than unity, mixed convection is important and we are falling out of ANTEO+ validity domain.

In order to ease the data reading from the following tables, the rows which represent cases in the anticipated validity range of ANTEO+ have been highlighted with a gray background.

ORNL-19. In Fontana et al. (1973) a 19 pins bundle, part of the fuel failure mockup (FFM) facility, with the aim of testing operating conditions of liquid metal cooled systems has been used. The selected cases for this bundle are reported in Table 10, where COBRA-IV-I-MIT (hereafter simply referred to as COBRA) results taken from Fricano (2012) are also included, when available. In Fig. 19 a visual comparison is given where the SC labels on the abscissa are



(a) $T_{ing} = 200^\circ\text{C}$



(b) $T_{ing} = 350^\circ\text{C}$

Fig. 17. Graphical representation of calculated and experimental bundle average temperature gain for the lead-bismuth experiment described in Litfin et al. (2011). Dots represent experimental data, while continuous lines represent ANTEO+ results.

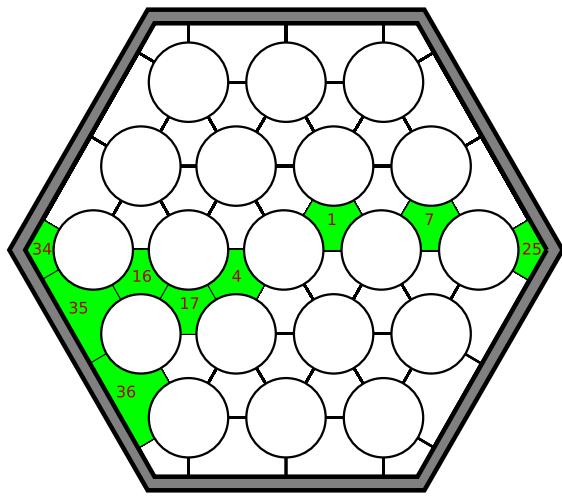


Fig. 18. Experimentally instrumented SCs and corresponding ANTEO+ numeration for the ORNL-19 bundle.

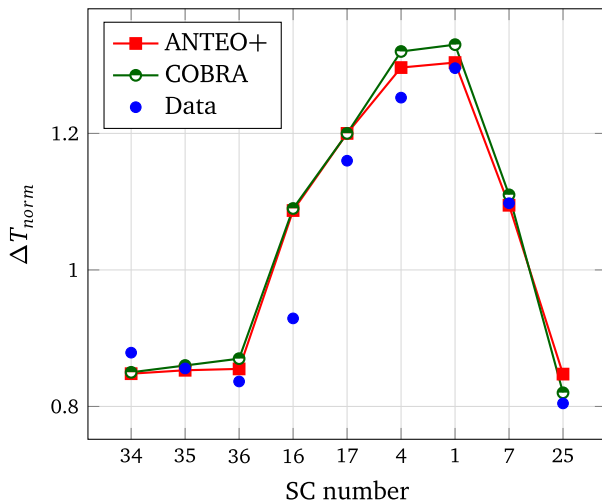


Fig. 19. Representation of experimental and simulations results for Test 2-Run 105 of the ORNL-19 bundle.

highlighted in Fig. 18; the ordinate is the normalized temperature gain expressed as

$$\Delta T_{norm} = \frac{T - T_{in}}{T_b - T_{in}}, \quad (55)$$

where T_{in} is the inlet temperature and T_b is the average bundle outlet temperature.

In order to keep this subsection clean and understandable, all the results presented in the validation of SC temperatures for hexagonal wire spaced bundles cover only the CTm correlation, which is the best among the ones introduced in Section 2.8.4 for this type of bundle. The other correlations results are summarized in Table 18.

Toshiba-37. In Namekawa et al. (1984) experiments performed on a 37 pins bundle used by the Toshiba Corporation Nuclear Engineering Laboratory in Japan are described. The bundle comprises three regions which can operate at different power levels, thus enabling transverse power skews to be simulated; moreover the axial power shape is a chopped cosine with an axial factor of 1.2. Three different skews, each one in three different flow conditions, were examined as reported in Table 11 along with the results of the simulations.

From Table 11 we can see that errors are higher for the skew cases; in Namekawa et al. (1984) an estimate for the heat loss through the duct is given as 2.5% of the power discharged to the

Table 11

Comparison summary between simulations and Toshiba-37 bundle experimental data.

Identifier	Power (kW)	Flow rate (kg/s)	γ	$\bar{\varepsilon}$ [%]	
				ANTEO+	COBRA
Power uniform					
B37P02	53.60	1.37	0.08	3.6	3.6
C37P06	41.02	0.31	0.71	1.3	1.5
E37P13	13.40	0.09	1.30	1.5	0.6
Power skew 1.40 ^a					
E37P17	53.50	0.93	0.15	4.8	5.7
F37P20	53.82	0.31	0.96	1.9	1.9
F37P27	32.56	0.17	1.40	5.1	4.2
Power skew 1.96					
G37P22	54.57	0.92	0.16	8.5	9.0
G37P25	54.57	0.31	0.97	5.6	4.6
L37P43	34.13	0.14	1.89	6.6	2.3

Table 12

Comparison summary between simulations and Toshiba-37 bundle experimental data when 2.5% heat losses are taken into account.

Identifier	$\bar{\varepsilon}$ [%]			
	Adiabatic		2.5% heat losses	
	ANTEO+	COBRA	ANTEO+	COBRA
Power uniform				
B37P02	3.6	3.6	2.5	2.7
C37P06	1.3	1.5	2.3	2.1
E37P13	1.5	0.6	1.5	1.7
Power skew 1.40				
E37P17	4.8	5.7	4.0	3.8
F37P20	1.9	1.9	2.7	3.2
F37P27	5.1	4.2	3.0	2.3
Power skew 1.96				
G37P22	8.5	9.0	7.2	7.7
G37P25	5.6	4.6	4.5	3.4
L37P43	6.6	2.3	4.3	3.4

SCs, which can partially explain the higher errors just mentioned. To test this hypothesis simulations with the bypass model explained in Section 2.10 were carried out adjusting the bypass flow in order to have approximately 2.5% of heat losses. The same simulations were performed in Fricano (2012) and results are outlined in Table 12.

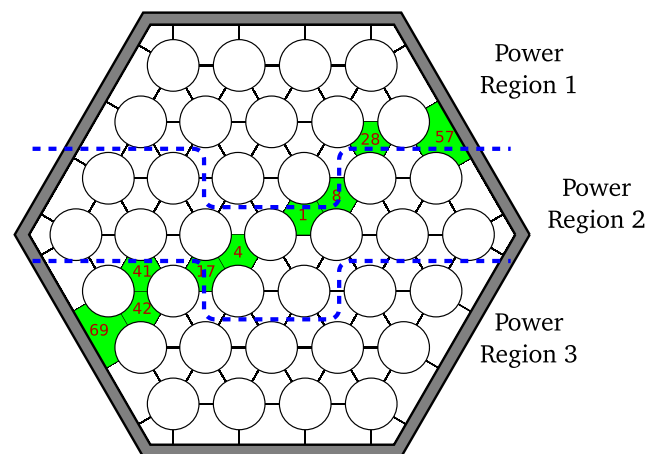


Fig. 20. Experimentally instrumented SCs and corresponding ANTEO+ numeration for the Toshiba-37 bundle.

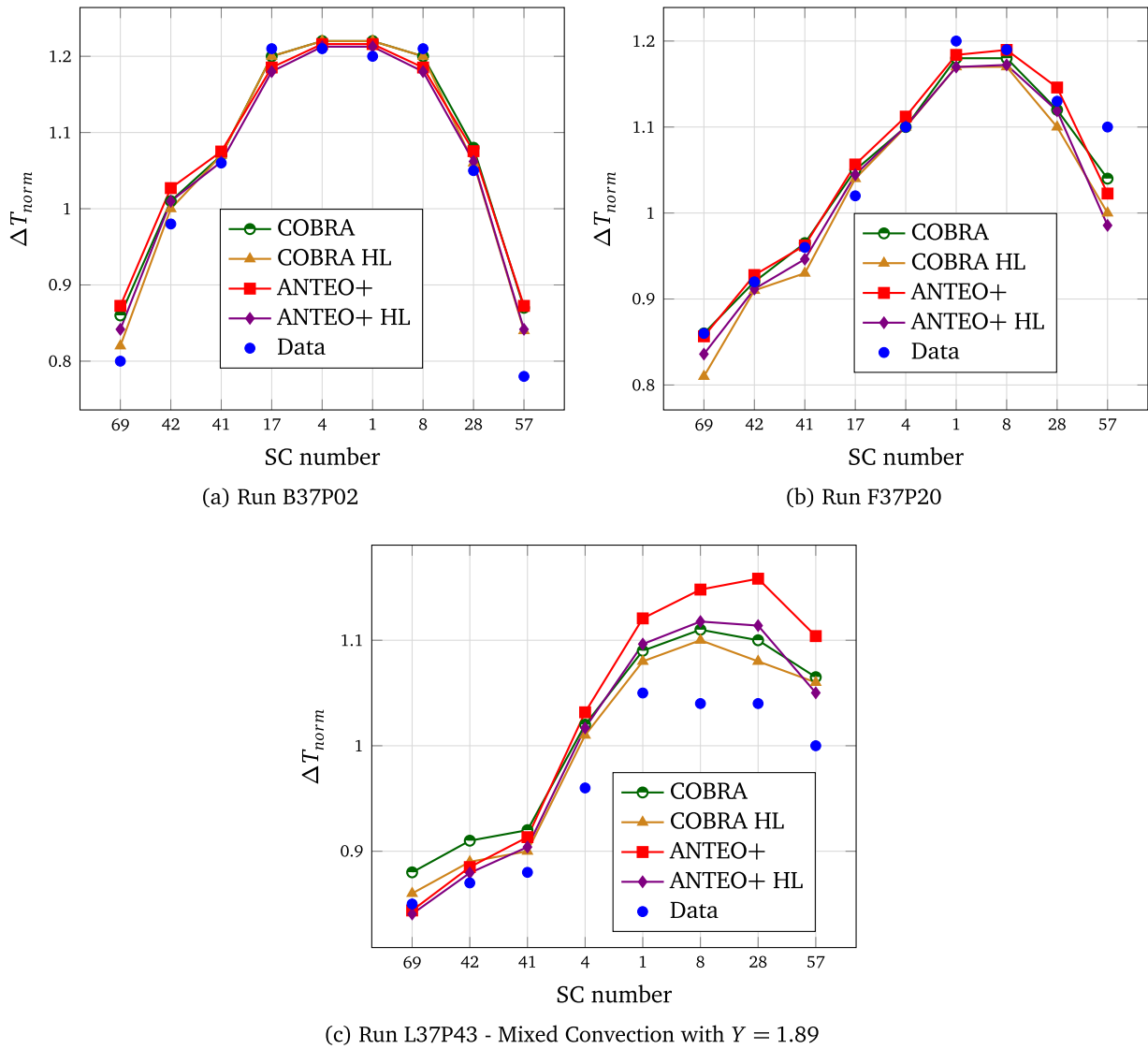


Fig. 21. Representation of experimental and simulations results for some illustrative runs of the Toshiba-37 bundle. In the plots HL means simulation with heat losses.

A visual comparison of COBRA and ANTEO+ can be found in Fig. 21 where the SC numbers are the ones highlighted in Fig. 20.

ORNL-61. The experimental bundle described in Morris et al. (1980) is geometrically identical to the ORNL-19 bundle, only the number of pins and the axial shape of the power profile (which is a chopped cosine with form factor of 1.38) are different. Measurements at three different axial locations were taken. As in the Toshiba-37 bundle the possibility of simulating power skews is present, but with a higher spatial resolution, since each row of pins has a modular power.

Results are presented in Table 13, where distances are taken from the beginning of the heated length, and visually

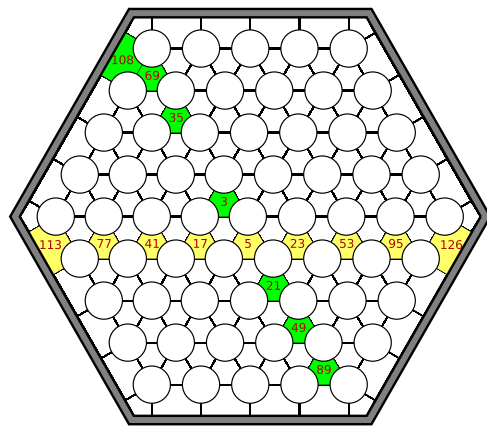
in Fig. 23 where the SC numbers are the ones highlighted in Fig. 22.

WARD-61. The experimental bundle of the Westinghouse Advanced Reactors Division (WARD) has 61 electrically heated pins (Engel et al., 1980). The WARD-61 bundle provides the largest collection of data of the present validation, thanks to the high number of operating conditions investigated. As the ORNL-61 bundle each row of pins has modular power and the axial power shape is a cosine with form factor of 1.4. Measurements were taken at a number of axial locations giving the possibility for validating the axial distribution along with the radial one; planes downstream the heated length are present; moreover, thanks to thermocouples placed in

Table 13

Comparison summary between ANTEO+ simulations and ORNL-61 bundle experimental data.

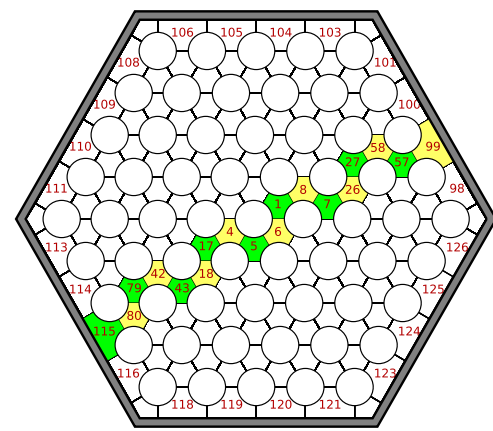
Identifier	Power (kW)	Flow rate (kg/s)	Y	$\bar{\epsilon}$ [%]		
				53.4 cm	63.5 cm	94.0 cm
Power uniform						
Run 1-4	1,220	8.490	0.06	2.9	4.5	2.1
Power skew 3.0						
Run 19-101	821.5	5.660	0.10	3.9	3.2	3.2



Measured elevations

63.5 cm, 94.0 cm 53.4 cm

Fig. 22. Experimentally instrumented SCs and corresponding ANTEO+ numeration for the ORNL-61 bundle.



Measured elevations

57.2 cm, 179.1 cm 115.6 cm

Fig. 24. Experimentally instrumented SCs and corresponding ANTEO+ numeration for the WARD-61 bundle.

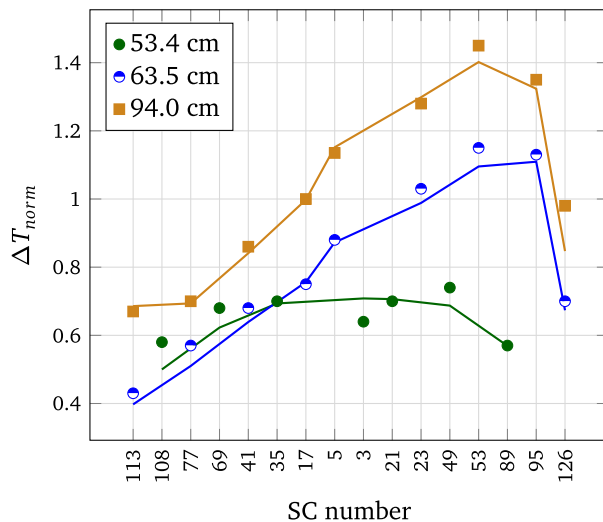


Fig. 23. Representation of experimental and ANTEO+ simulations results for the Run 19-101 of the ORNL-61 bundle. Dots represent experimental data, while continuous lines represent ANTEO+ results.

the edge channels, a detailed representation of the near wall temperature distribution is available for comparisons.

The selected cases are reported in Table 14 where runs from 1 to 6 are not official numbers; beside the uniform power case, three radial skews were simulated, where the one labeled 1.5:1:1.5 represents an U-shaped power profile. Simulations results are shown in Table 15 where the SC code ENERGY-II (Cheng, 1984) is included in the comparison⁷; distances are relative to the beginning of the heated section which is 114.3 cm long. A graphical representation of the results is given in Fig. 25 where SCs numbers are highlighted in Fig. 24. In Fig. 26 the near wall SCs temperature distribution is compared with ANTEO+, while in Fig. 27 the axial distribution of three SCs, comprising both heated and unheated lengths, is plotted along with COBRA results.

FETUNA-91. The last wire spaced hexagonal bundle analyzed is the mock-up described in Rameau and Menant (1984), which is an ensemble of 91 electrically heated pins. Only a uniform power

Table 14

Main parameters of the simulated runs for the WARD-61 bundle.

Run	Power (kW)	Flow rate (kg/s)	γ
Power uniform			
1	440.0	3.346	0.36
224	260.0	2.033	0.53
218	130.0	0.978	0.98
2	34.00	0.257	2.34
Power skew 1.5:1:1.5			
3	440.0	3.200	0.39
4	165.0	1.132	0.96
5	44.00	0.283	2.63
6	22.00	0.141	3.64
Power skew 2.0			
403	264.0	2.033	0.54
Power skew 2.8			
313	440.0	3.346	0.36
223	260.0	2.033	0.53
221	130.0	0.978	0.98
231	34.00	0.257	2.34
229	17.00	0.129	3.20

profile and mixed convection cases are analyzed. Results are summarized in Table 16 and visually in Fig. 28.

Hexagonal results summary. Given the high number of simulated cases it's worth to summarize the validation results just outlined for the hexagonal bundles.

The comparison with COBRA results is presented in Table 17 for both the mixed and forced convection cases.

The results for the whole ensemble of data points, including the correlations mentioned in Section 2.8.4 are shown in Table 18, where distinction is made between the two flow regimes. The errors reported are referred to the normalized temperature, but because the average temperature gain is around 100 K, the error on the true SC temperature gain (the numerator of Eq. (55)) is basically the same.

The cumulative distribution and the comparison with experimental results for the CTm correlation are reported in Fig. 29.

If we focus only on the SC with the maximum temperature gain (i.e. the hottest SC), which can be of particular interest for safety considerations, the error is considerably lower and equal to 2.7%

⁷ hereafter simply labeled ENERGY.

Table 15
Comparison summary between simulations and WARD-61 bundle experimental data.

Run	$\bar{\varepsilon}[\%]$								
	57.2 cm			115.6 cm			179.1 cm		
Power uni- form	ANTEO+	COBRA	ENERGY	ANTEO+	COBRA	ENERGY	ANTEO+	COBRA	ENERGY
1	-	-	-	5.3	13.3	-	-	-	-
224	2.0	2.5	2.4	2.4	3.9	2.6	1.1	5.5	2.2
224 edge	-	-	-	2.1	-	-	-	-	-
218	2.4	1.2	-	3.8	3.9	-	1.3	3.5	-
2	5.3	2.7	1.6	4.1	2.1	2.7	1.2	1.2	-
Power skew 1.5:1:1.5									
3	7.1	8.7	-	4.3	4.5	-	1.4	4.7	-
4	4.4	6.1	-	2.7	2.4	-	1.1	3.8	-
5	3.1	1.4	-	4.7	2.1	-	6.1	7.3	-
6	1.8	4.9	-	1.8	3.3	-	3.8	4.4	-
Power skew 2.0									
403	5.1	-	4.5	5.7	-	5.4	3.2	-	4.4
403 edge	-	-	-	6.0	-	-	-	-	-
Power skew 2.8									
313	3.7	7.3	-	2.8	15.8	-	3.5	16.2	-
223	3.0	10.9	2.5	3.8	10.2	3.6	2.9	14.4	3.7
223 edge	-	-	-	6.1	-	-	-	-	-
221	2.7	2.3	6.7	6.3	9.0	2.9	3.0	12.6	5.1
231	8.8	4.2	6.2	12.4	5.6	4.2	6.5	5.1	4.4
229	10.6	3.1	3.5	11.5	1.4	3.9	8.0	4.5	3.1

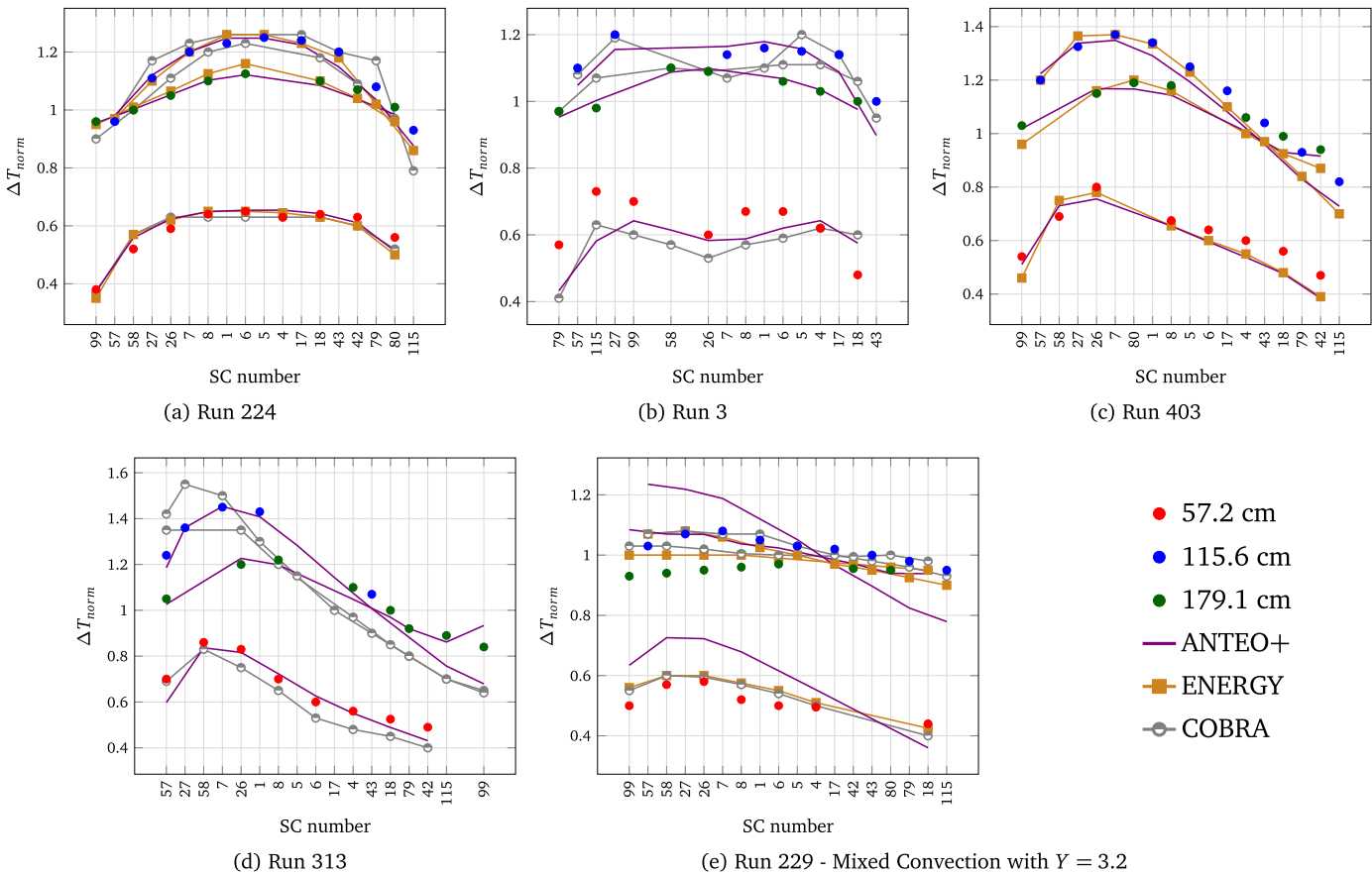


Fig. 25. Representation of experimental and simulations results for some illustrative runs of the WARD-61 bundle.

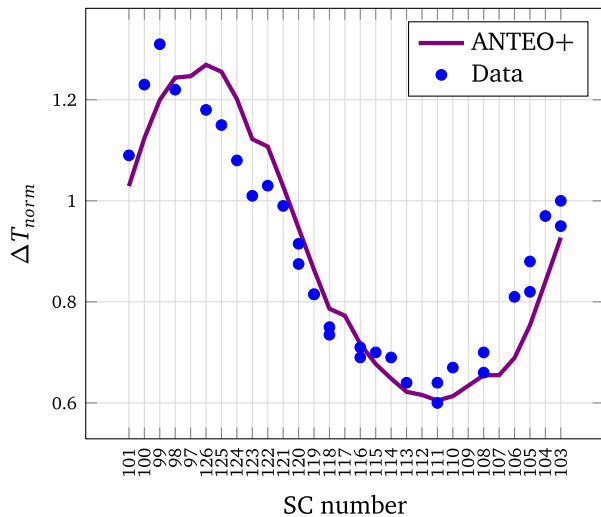


Fig. 26. Representation of experimental and simulations results for the near wall SCs temperature distribution at 115.6 cm for the WARD-61 bundle.

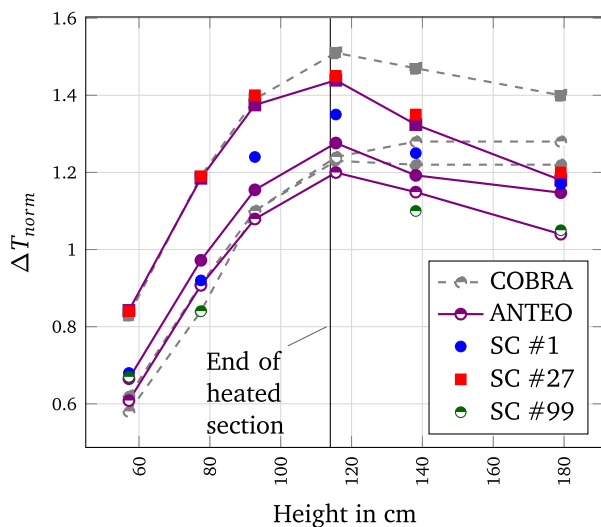


Fig. 27. Representation of experimental and simulations results for the axial SCs temperature distribution for the WARD-61 bundle.

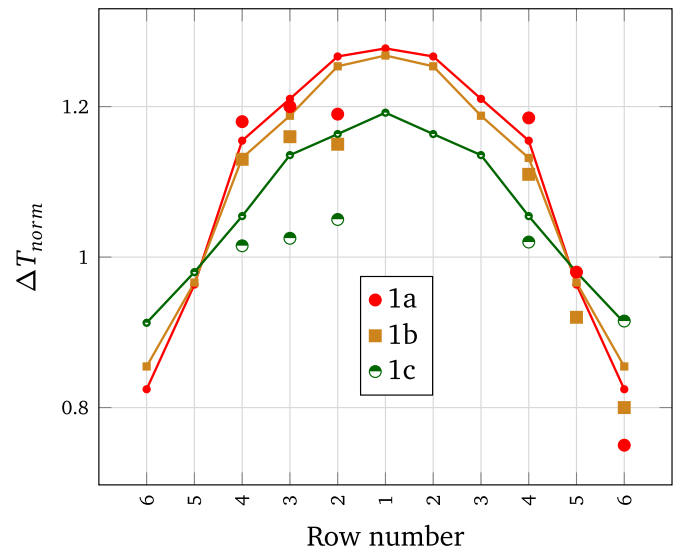


Fig. 28. Representation of ANTEO+ simulations and experimental data for the FETUNA-91 bundle.

Table 16

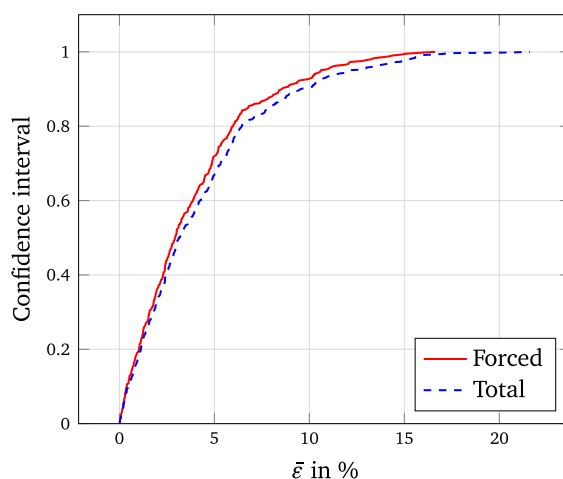
Comparison summary between ANTEO+ simulations and FETUNA-61 bundle experimental data.

Run	Power (kW)	Flow rate (kg/s)	Y	$\bar{\epsilon}$ [%]
1a	141.9	1.332	1.55	3.9
1b	71.34	0.676	2.33	4.3
1c	25.00	0.025	3.45	7.1

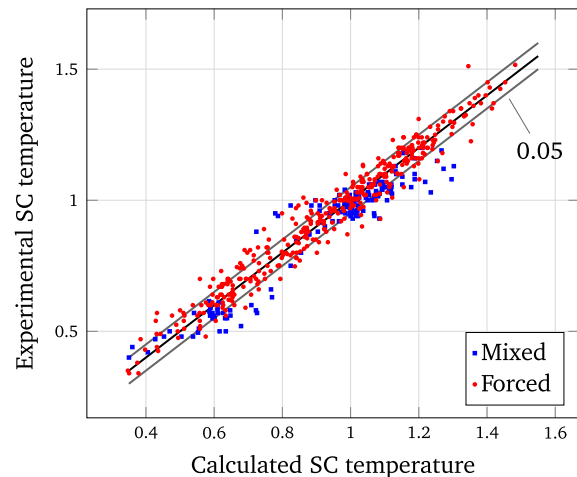
for the CTm correlation in forced convection; in mixed situations, instead, it is usually overestimated due to the higher flow redistribution in this regime.

Zhukov-25. For the validation, a peculiar square bundle, which relies on grid spacer for holding the simulation pins in position, has been included; the coolant, differently from all the previous cases, is a sodium-potassium alloy (22% sodium, 78% potassium).

During the 11th meeting of the International Association for Hydraulic Research (IAHR) a standard problem was presented by Zhukov et al. (2004) with the aim of assessing the reliability and accuracy of thermal-hydraulic codes. The model assembly is representative of the BREST-type reactor. The bundle has 25 pins with different diameters (see Fig. 30) and modular power capabilities,



(a) Cumulative distributions



(b) ANTEO+ simulated versus experimental SC temperatures

Fig. 29. Graphical results summary for the validation of the CTm correlation, for thermal mixing, in wire spaced hexagonal bundles.

Table 17

Comparison between ANTEO+ and COBRA average ε [%] for the hexagonal SC temperatures. Results are displayed by operating flow condition.

	Forced	Mixed
ANTEO+ ^a	3.3	5.5
COBRA	6.4	3.4

^a Only points where COBRA results were available are considered.

Table 18

Results summary for SC temperature validation on hexagonal wire spaced bundles. All correlations for wire-spaced bundles implemented in ANTEO+ are reported.

	CTm	Nijssing	Zhukov
<i>Forced convection (406 points)</i>			
$\bar{\varepsilon}$ [%]	3.9	5.6	4.9
90% confidence interval [%] ^a	8.6	10.8	10.0
<i>Mixed convection (162 points)</i>			
$\bar{\varepsilon}$ [%]	5.2	3.8	6.2
90% confidence interval [%]	12.8	8.7	15.1

^a 90% of the points have an error ε less than the reported value.

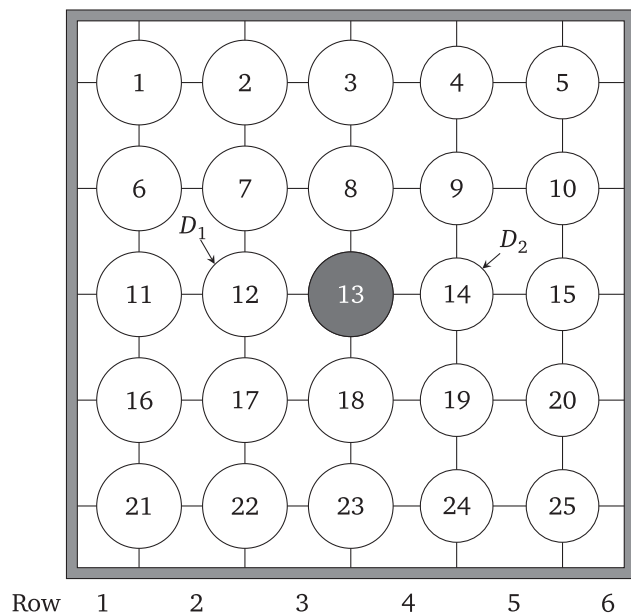
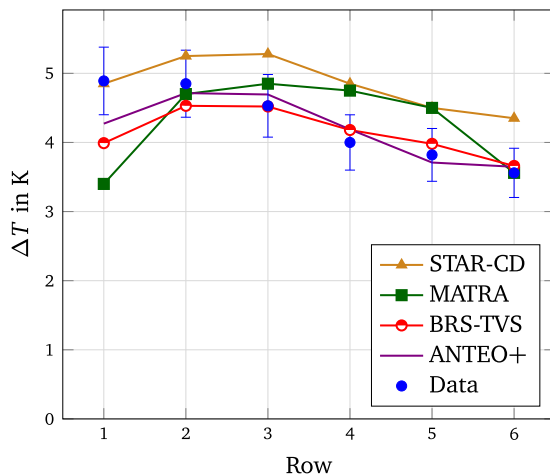
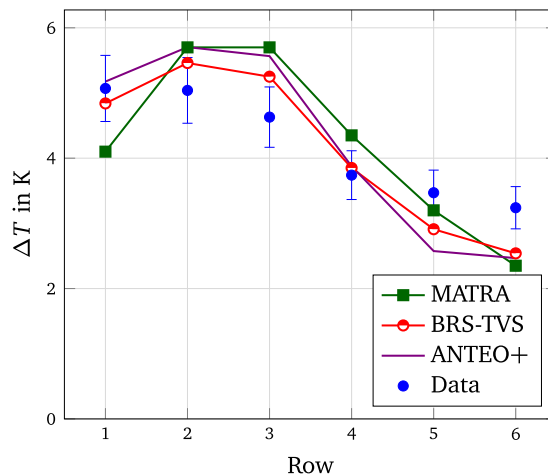


Fig. 30. Cross section of BREST-type model assembly used in the experiment by Zhukov et al. (2004). In figure $D_1 > D_2$ and the gray filled pin indicates the instrumented rod.



(a) Operating condition 2.



(b) Operating condition 5.

Fig. 31. Representation of experimental and simulations results for some illustrative operative conditions of the Zhukov-25 bundle. Results for STAR-CD (Carlsson and Wider, 2004) and MATRA (Son and Suh, 2004) codes are also included.

Table 19

Comparison summary between simulations and Zhukov-25 bundle experimental data.

Identifier	N_{15}/N_{10} (kW/kW) ^a	Y	$\bar{\varepsilon}_{rel}$ [%]	
			ANTEO+	BRS-TVS
1	1.35/2.00	1.35	10.3	13.0
2	1.65/2.00	1.49	4.9	6.1
3	2.00/2.00	1.69	7.2	5.1
4	2.00/1.65	1.58	13.2	10.3
5	2.00/1.35	1.46	14.8	11.2

^a N_{15} and N_{10} represent the power of the pin with diameter D_1 and D_2 respectively.

enabling different power skews to be investigated; in particular, there are two power regions that can be tuned corresponding to the two different diameters zones. The flow rate is fixed and so is the Reynolds' number, which is around 53400. In the simulations 200 axial nodes were used.

The benchmark was tackled by many specialists (Ohshima and Imai, 2004; Pena and Esteban, 2004; Carlsson and Wider, 2004; Son and Suh, 2004) with a variety of tools encompassing SC, distributed resistance and CFD codes.

Results, reported in Table 19 and visually in Fig. 31, were obtained using Kim's mixing model described in Section 2.8.4. The physical properties for the coolant were borrowed from Zhukov et al. (2004). In Table 19 the results of the BRS-TVS code (Afremov et al., 2004) are only reported, being the one with the overall best agreement with the data among the benchmark participants as reported in Zhukov et al. (2005); in Fig. 31 some of the other participants results are also included, in order to better contextualize ANTEO+ accuracy. The error bars reported in Fig. 31 correspond to an experimental uncertainty of 10% as suggested in Zhukov et al. (2004).

3.4. Nusselt's number

Once the SC temperature is known the Nusselt's number must be calculated in order to compute the outer clad temperature, which can be a tight design constraint. Some accuracy is therefore needed in order to reduce uncertainty margins and increase the degree of reliability of the overall core design.

For assessing the targeted accuracy of the correlations explained in Section 2.9 a comparison with experimental data has been carried out. Because the correlation by Mikityuk has been constructed

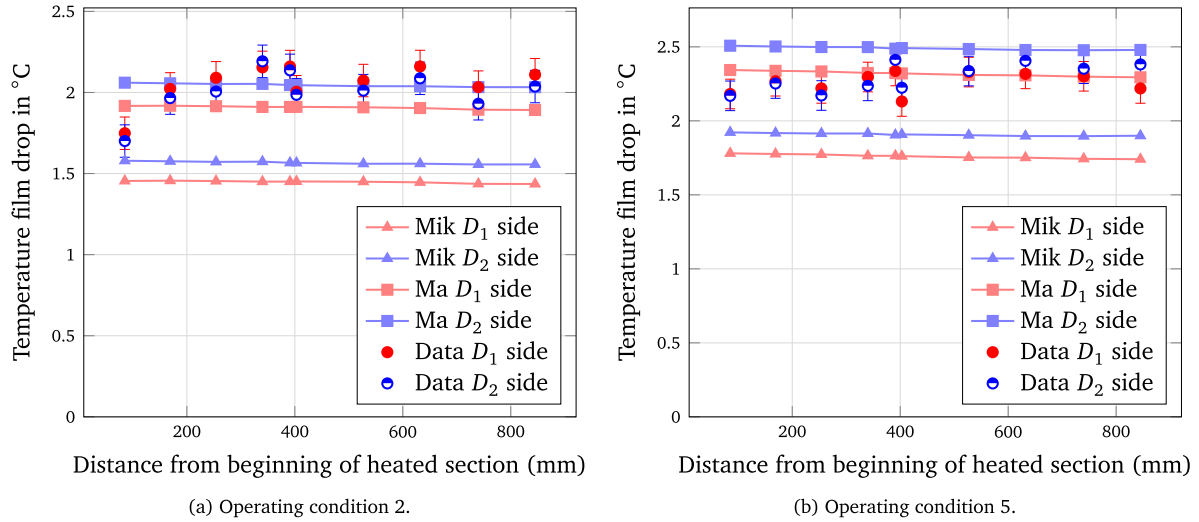


Fig. 32. Representation of experimental and simulations Nusselt numbers for some illustrative operative conditions of the Zhukov-25 bundle. Mik represents Mikityuk correlation, while D_1 and D_2 sides stand for the perimeter of the instrumented pin facing the side of the bundle with the respectively corresponding diameter.

Table 20

Comparison summary between simulations and Zhukov-25 bundle experimental data on Nusselt number for the instrumented pin in Fig. 30; errors relative to the clad-bulk temperature difference are given. D_1 and D_2 sides stand for the perimeter of the instrumented pin facing the side of the bundle with the respectively corresponding diameter.

Identifier	D_1 side		D_2 side	
	Ma	Mikityuk	Ma	Mikityuk
1	4.2	21.8	4.7	22.0
2	7.7	26.1	5.1	19.2
3	3.9	22.7	12.5	13.9
4	7.4	20.0	13.6	16.3
5	3.1	22.0	8.1	16.7

over an extensive database of hexagonal bundles and, recently, has been proven quite accurate in the lead-bismuth experiment described in Pacio et al. (2014), only a square bundle is analyzed.

The bundle in question is the one already presented in Section 3.3.2; measurements were taken around the colored pin in Fig. 30 and averaged over the two semicircles facing the different diameters regions; this process was repeated for 10 axial positions. Results for the five cases described in Table 19 are reported in Table 20 and graphically in Fig. 32. The error reported in Table 20 is referred to the difference between SC bulk temperature and clad outer temperature, because it is the quantity actually used in uncertainty analysis. Finally the error bars plotted in Fig. 32 are taken from Zhukov et al. (2004).

4. Discussion

The results presented in the previous section are critically discussed in this section, in order to find which recommended correlations must be used and the degree of accuracy to be reckoned to ANTEO+ inside and outside is anticipated validity domain.

4.1. Flow split

The flow split validation in the previous section covered the following range:

- $1.067 \leq (P/D) \leq 1.3$,
- $0 \leq (H/D) \leq 52$,

- $19 \leq N_{pin} \leq 217$,
- $320 \leq Re_b \leq 7.3 \times 10^5$,

which is deemed extensive and representative for ANTEO+ applications.

In Cheng and Todreas (1986) it is reported that the CT model can predict flow split parameters within 5% error. In our finding this is not true in the whole range, due to higher errors in the laminar region; in the turbulent region however, accuracy drastically increases, and the error drops well below the 5% limit. This is especially true for bare bundles.

The Blasius correlation gains some credibility only in the turbulent region and for bare or gridded bundles, as expected.

Given the presented results, the recommended correlation to be used in ANTEO+ is therefore the CT one. Blasius could be used in screening calculations, inside its validity domain, for evaluating complex geometric options.

4.2. Pressure drops

Because a thorough validation was already performed in Chen et al. (2014) for wired spaced bundles, only gridded bundles were analyzed, covering the following range:

- $(P/D) = 1.4$,
- $N_{pin} = 19$,
- $8.0 \times 10^3 \leq Re_b \leq 5.5 \times 10^5$,

which is not as extensive as the one used for the wired spaced bundles, principally due to the lack of experimental data for gridded bundles cooled by liquid metals.

Of the three correlations analyzed in Section 3.2 the one with the best agreement with the data is CT ($\bar{\varepsilon}_{rel} = 3.3\%$), followed by Blasius (4.6%) and Rehme (6.5%). The agreement for the CT model with experimental data is deemed acceptable and is the recommended one to be used in ANTEO+.

4.3. Temperatures

The most extensive validation has been performed on SC temperatures prediction due its prominent role in core design.

4.3.1. Bundle temperature

For the bundle average outlet temperature only a lead-bismuth eutectic experiment was simulated, with an error of 1.1%, proving the validity of the state equation implemented for this coolant (see Table 8). While sodium has been extensively used in the past, and so, some confidence on its physical properties has been gained, lead still needs further verification, even if lead-bismuth results are quite encouraging in this sense.

4.3.2. SC temperatures

The covered range in the SC temperatures validation is:

- $1.067 \leq (P/D) \leq 1.34$,
- $0 \leq (H/D) \leq 52$,
- $19 \leq N_{pin} \leq 91$,
- $0.05 \leq Y \leq 3.64$,
- $500 \leq Re_b \leq 7.3 \times 10^5$,

mostly for wire spaced hexagonal bundles cooled by sodium, with the exception of a square bundle cooled by NaK. Even if not complete, this range is still quite extensive and can be used as a guide to understand ANTEO+ targeted accuracy, at least in first approximation.

ORNL-19. For the ORNL-19 bundle various power skews were analyzed and we see from Table 10 that the error seems to increase at higher skews. The highest error is in the internal SC near the wall, possibly due to the inaccurate value of the mixing coefficient in this region; it must be said that being the power almost uniform in Fig. 19, SCs 7 and 16 should have similar temperatures, as predicted by the code, but not experimental data, giving an hint on experimental uncertainties in this bundle. The general agreement is however satisfactory with an average error of 3.5% and, for the analyzed cases, lower than the one obtained with COBRA.

Toshiba-37. As for the ORNL-19 bundle the error increases with the skew, while it decreases entering the mixed convection regime. The last effect is possibly due to the higher importance that heat losses have when buoyancy forces are not negligible anymore.

Results obtained by ANTEO+ and COBRA are quite in line for this bundle, with accuracy assessing around 4.3%.

ORNL-61. For this 61 pins bundle the power skew effect seems to be less important; this suggests that for bigger pins arrangements the spatial power profile is bound to be less determinant. The error remains pretty low for the two simulated cases and along the heated length, assessing around 3.3%.

WARD-61. A great number of cases was simulated for this bundle giving the possibility of better studying the effects of power skews and buoyancy on ANTEO+ accuracy.

It seems that for uniform skews the effect of buoyancy is less prominent due to the already flat temperature profile; increasing the skew however magnifies the effect of flow redistribution and the error grows larger (e.g. from 2.8% to 12.4% for the maximum skew analyzed).

The error drastically decreases far downstream the heated section, down to 2.2%, due to the mixing flattening effect and as before it is less sensitive to flow redistribution (5.1% in the mixed regime).

Comparing with the other codes we see that inside its validity range, ANTEO+ performs usually better than COBRA and in line with ENERGY. In the mixed convection cases the error increases and the other codes are able to better reproduce experimental results.

This bundle offered also the possibility of testing the near wall SC temperature distribution which was done with more than acceptable results (error around 4.7%) as shown in Fig. 26. Similarly the axial temperature shape has been investigated, as shown in Fig. 27, with some success. ANTEO+ is better able to follow the experimental shape compared to COBRA with accuracy respectively of 3.6%

and 8.2%. In particular, it can be seen that COBRA does not follow the data after the end of the heated length where the only non-zero term in the energy equation is thermal mixing. ANTEO+, thanks to its simplified and robust models, seems able to correctly reproduce the observed trend.

FETUNA-91. In the FETUNA-91 bundle only mixed convection cases were analyzed, and we see that, even with a uniform power profile, if the buoyancy gets strong enough, the error is bound to increase up to 7.1% and the maximum SC temperature overestimated.

Hexagonal results summary. Due to experimental uncertainties in measuring the SC temperature, in the energy balance, flow rate and the graphical acquisition of data, an error can be labeled as acceptable if it remains below 5%. With this discriminant we can see from Table 17 that ANTEO+ is more effective than COBRA in the forced convection regime (with errors respectively equal to 3.3% and 6.4%) thus confirming the code philosophy and modeling efficiency; in the mixed regime COBRA is able to handle flow redistribution and so the situation is reversed (with errors respectively equal to 5.5% and 3.4%).

From Table 18, where results for hexagonal wire spaced bundles are summarized for all the correlations implemented in ANTEO+, it is seen that in the forced convection regime the CTm correlation for thermal mixing is the closest to the analyzed experimental data, followed by Zhukov and Nijssing. In the mixed convection regime Nijssing performs better, probably because it usually overestimates mixing; since there are no physical basis under the success of Nijssing in the mixed regime, results must be taken with care.

As a final remark, it should be noted that in the analysis, the transition from the cold to the hot geometry was never taken into account, even if it can have some impact, especially for the near wall SCs. Simplified models of thermal expansion were proposed in literature (Pramuditya and Takahashi, 2013), but they are not always capable of improving results; for this reason ANTEO+ actually deals only with cold geometries.

Zhukov-25. The only square bundle analyzed used NaK as a coolant. We see that ANTEO+, using the correlation for thermal mixing by Kim, performs in line with BRS-TVS, which was the code closest to the experimental results as explained in Zhukov et al. (2005). Increasing the power skew increases the error because the experimental power shape is less sensitive to the skew than ANTEO+ (and BRS-TVS). This could be explained by errors in the flow split model, in the mixing coefficient or due to flow redistribution; for the latter in particular: even if Y is higher than 1 detailed studies indicate that it should not be dominant, suggesting that for gridded bundles the criteria for the onset of mixed convection is higher. Other possibilities are: the mixing coefficient is not high enough or the grid effect on the flow split is much higher than predicted.

Nonetheless, ANTEO+ accuracy for this bundle, assessing around 10%, has been proven to be one of the highest compared to other SC codes and CFD tools, especially considering the 10% experimental uncertainties suggested in Zhukov et al. (2004).

4.4. Nusselt's number

For the Nusselt's number only one square bundle was used as explained in Section 3.4. The experimental uncertainties for the temperature film drop are around 0.1 degree which is about 5%. With this in mind we see from Table 20 that Ma correlation performs better than the one from Mikityuk in square geometries, with errors respectively around 7% and 20%. We also notice that the entrance effect seems higher in the operating condition 2 than 5, giving an hint on the experimental uncertainties. Finally we see that the grid effect on heat transfer seems to be of the order of the uncertainties and so negligible in first approximation.

Table 21

Summary of estimated uncertainties and confidence intervals of the main SC parameters in ANTEO+ validity range. A reference experimental uncertainty is also given for qualitative comparison.

Parameters	$\bar{\varepsilon}$ [%]	90% confidence interval [%]	Experimental uncertainty [%]
X_i	4.8	11.4	5 ^a
ΔP	3.3 ^b	5.8	5
ΔT_{norm}	3.9	8.6	5
$T_{co} - T_b$	7.0 ^b	12.5	5

^a Flow split uncertainties vary with flow regime being around 5% for turbulent flow and as high as 10% for the transition region.

^b This is $\bar{\varepsilon}_{rel}$ [%]. Because the quantity to which $\bar{\varepsilon}$ refers to is close to unity, the difference between $\bar{\varepsilon}_{rel}$ and $\bar{\varepsilon}$ is basically negligible.

In ANTEO+ the recommended correlations are therefore: Mikiyuk's for hexagonal bundles and Ma's for square ones.

4.5. Validation overview

In the validation performed in Section 3 and discussed in the present section, basically all the models outlined in Section 2 were, at least preliminary, validated with the exception of the bypass model presented in Section 2.10. The validation has highlighted the capabilities of the models implemented in ANTEO+ and their interactions in reproducing experimental data.

The flow split model, especially in turbulent flow, is very accurate for both wire spaced or gridded bundles, with an accuracy of 1.8%.

Pressure drops can be predicted with some accuracy in basically every operative condition, even if validation is still needed for gridded bundles, where the error is 3.3%.

SC temperatures are calculated with an accuracy around 3.9% in a variety of flow conditions and geometric arrangements. The validity range of ANTEO+ is the single phase forced convection regime and an index has been used to warn the user if this regime has been breached; in this case the accuracy of the code can be reduced (down to 5.2%) especially for small bundles with high power skews. In the peculiar bundle Zhukov-25, some of the new geometric capabilities outlined in Section 2.11 were used proving the ability of the code in handling new configurations in the aforementioned validity range, with an error assessing around 10%.

The Nusselt's number has been also validated in square geometry, while the validation of the hexagonal one was borrowed from literature. The correlation implemented seems to be able to reproduce experimental data, with accuracy of 7%, but further validation must be carried out in order to quantitatively assess the targeted accuracy, especially in square geometry.

A summary of the estimated uncertainties associated to the main SC parameters, in ANTEO+ validity range, is reported in Table 21 along with the reference experimental uncertainties in order to contextualize our validation results. It must be stressed that these are specific reference uncertainties of the data used in the validation, but typical values found in literature can differ (e.g. uncertainties in clad-bulk temperature difference usually are higher than the one reported in Table 21).

5. Conclusions

In the present work the SC code for thermal-hydraulic analysis of LMFBRs, ANTEO+, has been extensively outlined. The main objectives of ANTEO+ are simplifying the problem description without penalizing accuracy thus enabling a more transparent interface with the user, and having a clear and identifiable application domain. In order to achieve these aims, a validity range has been decided beforehand, namely, the single phase forced convection regime.

The models and correlations implemented in the code have been presented stressing their expected validity domain. These expectations have been tested during the validation phase of the code, finding general agreement with experiments. In particular, a thorough validation has been performed on the models for flow split, pressure drops, SC temperatures and Nusselt's number quantifying the accuracy targeted by ANTEO+. Moreover, comparison with other codes and tools has been performed revealing the generally high level of ANTEO+ accuracy in its validation domain. When falling outside this domain however accuracy can drastically decrease and results start to become excessively conservative.

For this reason, future work can be focused on the extension of the validity domain to mixed convection because it can be important for heavy metals. Lead, for example, has a strong tendency towards buoyancy and falling in the mixed regime is possible even at relatively high Reynolds' numbers. Another possible improvement might be the creation of a model which accounts for the transition between cold and hot geometry.

Concluding, ANTEO+ is here presented as a robust subchannel code for application in single phase forced convection regime, which enables the user to more easily follow the calculation process and thus facilitating the results interpretation phase, including the identification of the confidence interval of the results.

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